Corncob Redux - Other Worlds Campaign

Lost Engines - Corncob Redux
Reviewed edition of the original game documentation.
Controls and changed rules describe Redux defaults.
Original credits, source tables and archival details are retained.

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Getting Started

Thank you for purchasing the Corncob Other Worlds Campaign!

The game is easy to install and use. To install Corncob, insert the distribution disk into your computer's drive. Type INSTALL, and press <ENTER>. Answer the questions on installation drive and directory. After the installation program is finished installing the software, type CORNCOB to begin the game.

If you just want to get in the cockpit and fly, then press <ENTER> at the main menu. Doing so will bring up the Training Missions menu. Just press <ENTER> again, and you will receive the pre-flight report, which gives information about the missions you fly. You can exit the pre-flight report with <ENTER>, and you will be in the cockpit.

You may get help for the various submenus by pressing I. For help inflight, please use the F4 key. If you are having difficulty flying and turning, please read the documentation.

Title and credits

[Historical DOS documentation - retained for reference]

Corncob 3D
The Other Worlds Campaign

A VGA Air Combat Game
by
Kevin Stokes

An MVP Software Production

For IBM and compatible computers
1992 Pie in the Sky Software
Printed in the United States

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The Story So Far

The date is August, 1949. This is not the 1949 that we remember, though. In this 1949 WWII never happened (this is because among other factors, Adolf Hitler was killed at the age of 19 when hit in the head by a thrown beer bottle), but all was not peaceful in this alternate age.

In 1938 aliens swept down out of the skies and began to set up structures. Although many attempts were made to communicate, the aliens did not respond. It soon became obvious that these creatures simply were not interested in us, or in any other life on the planet. When human dwellings or sites were in their way, the aliens destroyed them and continued their seemingly random construction plans.

So in 1941 the world governments agreed to band together in an alliance for humanity. No attacks were launched until November 1941. The plan consisted of a simultaneous all out blitz against the extraterrestrials in every part of the globe. The battle was waged for over four years. Finally the effort succeeded and the aliens were driven from earth.

Now, however, they have set up residence on other planets in the solar system. Clearly they are planning another conquest of earth. Once again the world's governments are banding together, but this time to drive them from the solar system.

Your mission is critical in this plan. You must go to other planets, destroy several alien structures, and in the process, many aliens themselves. If you do not succeed the entire human effort will fail.

Hardware Requirements

[Historical DOS documentation - retained for reference]

Hardware Requirements
  Corncob requires an IBM PC/AT-type computer with a speed of 12Mhz or higher, a VGA color monitor, a hard drive, and 570K free RAM. A joystick
is recommended but not required for successful game play.

  For music and inflight sound, you must have an ADLIB compatible sound card installed in your computer.

What to do if Corncob Crashes Your Computer
  Corncob has been tested on a variety of systems and has run reliably with no crashes for weeks of game play. However, Corncob does do some rather unorthodox things that may cause problems with certain TSR programs.

  If you have a problem running Corncob on your system, reboot your computer from a floppy disk, with no TSRs or other software loaded.

  NOTE: If Corncob fails during loading, please delete or rename the file
CCTITLE.EXE, and then let us know about it. This will prevent the animated title screen from running.

Redux Controls

These are the default bindings. F4 opens the current bindings during flight; Enter selects a binding to change it. Escape or P opens the pause menu, including display and control options. Delete clears a binding during capture; Escape cancels capture. Keyboard and mouse bindings can coexist: capturing a key changes the keyboard binding; capturing a mouse button changes the mouse binding. Delete clears both.

Flying
  Arrow keys or keypad 4/6 and 8/2: roll and pitch.
  Z / X or keypad 7 / 9: left / right rudder.
  + / - or keypad + / -: throttle up / down.
  Period or keypad 0: hold wheel brakes.
  Space or left mouse button: machine guns.
  Left Shift: heavy cannon. C or right mouse button: missiles.
  S: toggle sound effects. A: autopilot. F: flaps. I: rocket booster. B: drop bomb.
  E: step out behind the wing, or eject if airborne.
  R: rear view. L: cockpit. Num Lock: look mode.
  Hold Ctrl with arrow keys to look around.

On foot
  W / A / S / D: forward / left / backward / right.
  Mouse: look around. Space: jump immediately.
  Left Ctrl: drop through a supporting platform.
  Left mouse button: pistol. Q: open parachute.
  Semicolon: plant assassin charge. B: arm/detonate charges.
  V: call rescue. Caps Lock: remote control of your aircraft.

Shared controls
  T (hold): tenfold fast-forward while flying or walking.
  M: map. O: toggle radio messages. G: ground detail.
  P / Escape: pause menu. F1: boss screen.
  F2: intelligence. F3: current score. F4: control bindings.
  F6: cockpit / first person. F7: third-person view.
  Mousewheel: scroll menus, documentation and tower briefings.

Joystick buttons 1 and 2 fire guns and missiles during flight.
Modern joysticks do not need the DOS calibration controls.

Fast-forward advances the whole simulation, including enemies, projectiles and production. They do not sleep while you hold T.

Menus, documentation and the tower
  Arrow Up / Down: select or scroll; mousewheel also scrolls.
  Enter: accept. Escape: back/cancel. Backspace: erase text or go back.
  Home / End: first / last entry or beginning / end of a document.
  Page Up / Page Down: scroll documents by a page.
  Left / Right: adjust options; switch world groups in the world list.
  Insert: toggle detailed pilot records. Highlighted letters: shortcuts.
  Type a chapter number in the documentation contents, then Enter.
  In the tower, Enter selects a highlighted mission and closes the briefing; if airborne, this relocates your aircraft to that mission.
  Tower aliases: J/K or 2/8 scroll down/up; D/U move half a page; Space pages down; G goes to the start, Shift+G to the end; Q exits.
  Ctrl+F/N and Ctrl+B/P scroll; Ctrl+Y pages down, Ctrl+O pages up; Ctrl+T goes to the start, Ctrl+R to the end; Ctrl+H exits.

Redux Controls (continued)

Map mode
  Arrows or keypad 4/6 and 8/2: move the map viewpoint.
  Page Up / keypad 9: descend (zoom in).
  Page Down / keypad 3: climb (zoom out). Hold S for slower zoom.
  Home / keypad 7: reset the map to your position.
  Hold Left Shift with arrows: rotate the map viewpoint.
  Escape, Enter, Space, or the configured Map key: return to flight.
  Map mode pauses the sortie; looking at a map does not advance enemies.

Mission builder (fixed editor keys, separate from flight bindings)
  In the world list, B opens the selected world in the builder.
  Arrows: turn the view; keypad 5/0: move forward/backward.
  Page Up / Down: lower/raise the viewpoint. 1..8: movement speed.
  Left Shift: show cursor or commit object; Left Ctrl: hide cursor.
  C: create an object. Space: select a nearby object. D: delete it.
  + / -: next/previous object type.] / [: next/previous quality.
  Hold Left Alt with arrows and Page Up/Down: move cursor/object.
  Hold A with arrows: rotate selected object.
  F9: toggle special flag. F10: toggle objective-complete flag.
  Caps Lock: level the view. Alt+H: remember cursor height.
  H: move cursor to the remembered height.
  B, M or Escape: switch overhead/3D view. F5: save and return.
  Edits are exported under custom_worlds in the Redux save directory.

Other retained shortcuts
  Ctrl+C during flight: abort the sortie without awarding a score.
  Alt+I: enable invulnerability; in Other Worlds this toggles it.
  Escape stops a recorded demonstration without saving it.
  While a radio message is visible, O disables future messages;
  any other non-modifier key dismisses the message.
  F1's boss screen dismisses on any non-modifier key.

Mouse sensitivity and inverted look are adjustable in Options and
controls. Losing focus releases the mouse and pauses the game.
FOV, draw distance and ground detail do not affect simulation speed.
Ground-detail range changes how widely the dots are spread. A wider
range lets them shrink before the field ends. Increase the point count
as well if you want more nearby dots. The default range is eight times
the DOS field; choose 1 for the original extent.

Keyboard Play and How to Handle the Rudder

A great deal of effort was put into Corncob to make keyboard play work well. The keyboard is interrupt driven, and your key presses are timed to within 1/1000th of a second. Also, Corncob has no problem detecting keystrokes even when 3 keys are held down at once.

The keys that even joystick players have to contend with are the rudder keys. The problem is how do you command just a little rudder? If you hold the key down the plane will go into a heavy sideslip, and eventually stall. The method of varying the amount of rudder desired is by rapidly pressing and releasing the key. The average amount of time in which the key is down communicates to the game how much rudder is desired.

This will take some getting used to, but when you get used to it, it becomes second nature. To operate the rudder, the best finger position leaves the left hand middle finger on the z key for left rudder. This leaves the right index finger on the x key for right rudder. Now your 3rd finger is well positioned for pressing the left shift key down for cannons (use Period or keypad 0 for brakes). For lefties, the rudder controls are 7,9 on the keypad.

For the keyboard player, the plane must be controlled from the keypad. Put your middle finger permanently on the 5 key. Use this middle finger for both the 2,8 nose up/down keys, and use the index and 3rd finger for your left/right roll keys. It's very important always to use the above fingers for best performance. The reason is that in Corncob, you frequently must use more than one arrow key at once to perform a maneuver.
If you peck the keys with just one finger, your ability to react quickly and use combinations will be limited. It's best to force yourself to do it the above mentioned way, or else you will be stuck at the harder missions that require more precise flying.

Remember that to apply gentle tweaks to the control, you must press keys on and off rapidly, with the average time spent down being the desired roll/pitch rate. You have three displays on the front panel that show you what you are presently sending to the control stick. These meters are just to the right of the ALT light.

Don't forget to use the rudder with the left hand. Using the rudder can really help you pick off targets without having to bank and pull back on the stick.

NOTE: If the arrow keys suddenly appear not to be functioning properly, perhaps you have bumped the numlock key. Numlock chooses the mode in which your keypad keys operate between control/views. Note that the numlock light will not change state as you switch between views, and control.

Training Missions

If you don't want to fight in an actual theater; i.e., you just want to hop in a plane and shoot at aliens, then the Training Missions are for you. Use the menu to specify the number of enemy KLA sites, AAA batteries, and ground vehicle sites, and you are ready to engage in combat.

You will be placed in a universe that is empty except for your airstrip and the enemy field of operations. The basic setup of the enemy territory will always be the same. There will always be an orb protected by forcefields, and 15 fuel storage facilities. The fuel storage canisters (or Ticks, as the pilots call them) are always the primary mission objective.

If you are flying training missions with a non-trainee pilot, be aware that your pilot scores will be affected by your performance on your flight, and your pilot can be just as easily killed or captured as when you are flying in a theater.

This is why it is important to remember to fly the training mission as a trainee if you just want to bomb around.

Combat Missions: Theaters-of-Operation

To fly a mission a pilot first chooses in which theater of operations he wishes to fly. A theater of operations is a terrain filled with airports and areas of enemy concentrations. After opening a theater, select F from the combat missions menu in order to fly a mission in that theater.

A pilot selects his mission by either flying there from the airstrip or simply pressing F2 to bring up a list of the local missions. The pilot selects the mission by using the arrow keys. Then by pressing the <ENTER> key, he/she will be teleported right to the action.

After you have completed your mission you must fly back to your airstrip to get credit for your victory. If you can land the plane, your points will be much higher than if you quit during a flyby.

Still, even if you cannot land the plane safely, quitting the flight via the <ESC> key must be done close to airstrip to avoid losing a plane. The definition of close to your airstrip is when your airstrip's radar dish is blinking.

Most theaters of operation have nine airbases from which the pilot can take off and in which he can land. Each airbase has intelligence information about the enemy forces in the vicinity. Intelligence info can be gotten from entering the control tower on foot or by using your radio (F2 key) when in the air. The information you receive will always be from the last airport you were in.

If you wish to get intelligence info from a different airstrip, simply do a flyby over the tower of the new airstrip. Then the next time you use your radio via F2, you will get the intelligence about the missions surrounding that airstrip.

For example, suppose you started a combat mission from the RDU airstrip.
Once you hit F2 you realize that you have already completed all the missions here, and you would like to fly the missions at the Crickety Point airstrip. So you simply fly due North until you make a sighting of the Crickety Point airstrip. Then you make a flyby of the tower, and as a result the Crickety Point radar tower starts flashing. Now when you hit F2, your intelligence report will be from Crickety Point.

One of the most important things about theaters is that they have a history capability. This means that anything a pilot has destroyed will stay destroyed forever. For example, say that yesterday a mission was flown in which the Devil's Aquarium was demolished. If that same theater

Combat Missions: Theaters-of-Operation (continued)

is flown in today the wreckage will remain. This applies to friendly objects as well as enemy objects. Your goal in each theater is to accomplish all the mission objectives listed in the intelligence reports.
Typically this will take many missions, and on your last few missions the
theater will be filled with destroyed alien targets. If you use up all your planes, you will have to open a new version of that theater (with all
alien bases intact) and start again from scratch.

This brings us to the topic of plane conservation. Planes are a precious
resource that should not be wasted. Each theater has only so many planes
parked at the various airstrips. If a pilot is not careful and loses too
many planes via ejection, then the unfortunate pilot will find himself with no airplanes left in the whole theater.

The method to get the most points, and the way to get through a theater,
is by nearly always nursing planes back to the nearest airstrip.

Even a heavily damaged plane can be repaired simply by landing and strolling over to the control tower on foot. If the pilot cannot land, then ending the mission with a low flyby will assume a safe landing with plane intact, and the next time the game is played the plane will be repaired. Please note that even a crashlanded plane can be repaired if the wreckage is close enough to the airstrip tower.

If desired, some older model planes can be brought out of storage, but this should be used only when desperate. These brown planes have reduced
performance specs. The option for bringing these planes out of storage is
only available when you attempt to fly a mission from an airstrip that has
no airplanes available. Of course, if a pilot runs out of planes in a theater, or would like a fresh start, there is always the option of reopening a new theater. A pilot can have up to 15 theaters open at one time, so he may fly one mission in MILKRUN theater, then fly a mission in
FLATLAND, and so on. Closing an undesirable theater does not affect the pilot's record. In fact even if a pilot is killed in action, he can be resurrected. The only cost of resurrection is the fact that the computer
keeps track of how many resurrections you did and places this number by your name in all records.

Each pilot has records kept both for his total career and his performance in each theater in which he has flown.

Friendly Objects

Airplanes
  Every airstrip contains one or more airplanes you may fly. To get into
a plane, simply walk up to it and enter. To get out of a plane use the e
key. There are three different plane types in the game.

  The blue-winged planes are the standard front-line ground attack planes.
 The brown-winged planes are older planes that have been brought out of storage in the case of an emergency. These planes have less engine power.

  Lastly, there are experimental black flying-wing planes. These planes use the latest stealth technology, and there are some drawbacks. First, these planes are extremely heavy, so the ground crew cannot move them. When you fly the missions you must find the plane yourself, and it should
be wherever you left it. Also note that these planes will fall like a rock when stalled. You will have less than half the time to bail out before hitting the ground in these aircraft.

  The high speed of these planes is actually faster than the initial speed
of their missiles. When you fire missiles you will see them fall behind in
the rear view. But depending on your airspeed, they may pass you as they
accelerate towards the target. Your gun tracers will appear to be strangely affected also. This too is a side effect of your great speed.

Control Tower
  Every airstrip has a control tower. This tower contains the radio equipment and intelligence officers. If a pilot enters the door of the control tower on foot, his plane will be repaired if close enough.

  If a control tower is destroyed it may be impossible to get intelligence
on future missions.

Radar Unit
  The radar tower watches for your safe return to home base. If you want
to have your plane on the next mission, you must make sure the radar tower
is flashing before ending the mission with the pause menu's End current sortie option. The radar
tower begins flashing when you get to within a certain distance of the control tower.

Ending the mission with the radar tower flashing makes a big difference in
the computed pilot score.

Runways
  Runways are the ideal location for taking off and landing. Your plane can also do both on the grass, but the risk of crashlanding is much

Friendly Objects (continued)

higher
when not landing on the runway. Taxiing on the grass bounces the plane about and can cause a crashlanding even at low speeds.

Rescue Van
  When a pilot is downed behind enemy lines he can be rescued by the rescue van. He must first eject from the plane and then use his handheld
portable radio to call for assistance. (Press the v key.) You may only call the rescue van when standing on the ground. The handheld radio will
not function in your plane or while floating to earth with your chute.

  When called, a rescue van will be dispatched and will appear shortly. The van will approach you and stop when within a few feet of the pilot. When the pilot enters the van the mission is over, and the pilot is considered home safe.

  The rescue van is not indestructible. The van will be destroyed if it collides with another object on its way to pick you up. It is up to the pilot to maneuver himself so that there are no obstructions between the approaching van and himself.

If the rescue van crashes you must call another one.

Spies
  Some missions require you to pick up information on foot from spies. Spies are people who wear all black. They give you the thumbs up when you come close enough to grab the info. Consider your mission objective complete when you get this hand signal from the spy.

Enemy Defenses

Force Fields

Force Fields are passive defense systems used to protect resources by shielding them from firepower, and preventing our aircraft from flying through. Some forcefields shimmer, others are solid colors. Some fields appear as a rectangular frame that may or may not vary in size with the passage of time. The solid and shimmering force fields require the most power and therefore only click on when needed. When a bullet or missile, or a plane itself comes near a solid forcefield, the forcefield will pop into existence. Almost all bullets and missiles are stopped by the forcefields, as well as bombs. Do not use missiles when your plane is too close to field, as they will explode in your face when launched. A forcefield will cause damage to a plane at a certain rate during the time in which the airplane is flying too close. If a forcefield must be flown through, then to minimize damage, fly through perpendicular to the surface, and the higher the airspeed the better.

Note that the shimmering solid fields are in fact opaque, even though the optical illusion makes them look translucent.

The force fields that vary their size and shape with time stay on at all times. The boundary of deadliness of these fields moves with the field's image. You can use this fact to eek in a few shots at protected targets.

In almost all cases forcefields are powered by a local generator orb. Therefore when you take out the orb, all the local forcefields will drop.

Note that the enemy's fire can be stopped by forcefields also. A favorite technique of our pilots is to lure KLAs into a nearby field. Also AAA shells are stopped by FFs, so sometimes they can work to our advantage.

Please note that although forcefields will damage your plane, they will cause no damage to organic flesh, and in fact all forcefields can be walked on with no ill effects.

Barrage Balloons

Barrage balloons are huge membranes filled with a lighter-than-air gas. They float up and down randomly. Their purpose is simply to be in the way. They are organic and self-regenerating, so although they stop all our firepower, they themselves are indestructible.

Your plane can fly through them, but at a cost of many damage hits. BBs never go higher than a couple thousand feet. Their colors show something about the speed at which they rise and fall.

Enemy Defenses (continued)

Bee Swarms

Bee swarms are very similar to the barrage balloons discussed above. They float up and down, but they are much harder to see, and it is more difficult to determine their exact position due to their indistinct appearance.

AAA Batteries

These are one of the most threatening of the enemy's active defenses. AAA sites track your plane and send up volleys of explosive shells that burst in the air around you. You do not have to be hit directly to take damage from an AAA shell. The shells are designed to explode into fragments that tear through your plane, even if you are hundreds of feet from the explosion.

The AAA batteries vary wildly from site to site. We have found that there is a correlation between muzzle color and deadliness. Usually the brown AAA is the least threatening, while the white AAA is devastating.

The AAA quality value affects several capabilities. The lower quality sites cannot turn fast enough to track your plane if you are flying past.
Also, quality affects the maximum firing rate, the accuracy of the shells,
and whether or not the AAA has the intelligence to lead the target.

Note that AAA shells can explode KLAs. In the early days before the leading technology AAA sites were developed, the classic method of defense
against deathballs was to fly past AAA sites at a certain range. All the
shells would fall behind your plane to where the chasing missiles were.

Note also that ground-mounted AAA sites cannot fire horizontally. If they did, the AAA batteries would be destroying their own forces. You can
use this to your advantage, if you have the ability to fly extremely low.
The closer you are to an AAA site, the lower you must fly.

KLA Deathballs

These are enemy missiles that lock on to your plane and attempt to explode when close enough. They are launched either from missile sites or
from saucers. These missiles have varying abilities which you can determine from their color.

When a KLA becomes active, it will show up as a green dot on your radar
screen. When you see one visually you will notice that when it is pointed
at you it begins to flash. At this point the KLA has a lock on you. Most
KLAs will accelerate forward when they get a lock.

Note that deathballs always head directly towards you, and do not lead your plane, like AAA batteries do.

Enemy Defenses (continued)

When a deathball is close to you, you will hear a strange thrumming sound. The higher the pitch, the more deadly that KLA is. The louder the sound, the closer the KLA is to your plane.

Generator Orbs

Generator orbs produce power for force fields and normally act as a central structure to each concentration of enemy activity. The process used by the aliens to generate power has a strange disorienting effect on humans when close to the orb. Usually this effect decreases with distance, but under certain circumstances can become independent of position.

The triangular base of the orb is indestructible, but the ball on the top is not. When attempting to destroy it, aim for the center of the flashing ball. These structures are tough. It takes maybe ten missiles or two bombs and a few missiles to destroy one. When an orb is destroyed all the force fields in the immediate vicinity drop. This makes them of extreme strategic importance on some missions. In some situations it is wise NOT to destroy the orbs, since sometimes the aliens' force fields can be used to our advantage.

There is a special kind of orb that has an tractor beam that exerts a pulling force on your plane. This force is so strong that your plane can be pulled backwards through the air in some cases. This effect decreases with distance, so be careful about flying too close to this type of orb.

Some orbs have been designed to be invincible to normal weapons, and the only effective way to take them out is via suiciding the plane into them.
Make sure you bail out before contact, or else you will be killed in the ensuing explosion. If you must bail a good way away from the target, hit Caps Lock to control your plane remotely.

TRFRU

Tetrahedral Radio Frequency receiving units, or TRFRU, are antenna systems used for spying on all human military communications. They look like small black tetrahedrons spinning in the air. Guns are the weapon of choice for taking out tofu (tofu is the unofficial name). Be especially careful when attacking suspended tofu. They are so small that many pilots wait too long before peeling off to the left or right. This can result in a disastrous collision.

Mortars

Mortars are simple devices that are mounted on the ground and spew incendiaries when any human is near. They are used to deny low altitude flying.

Mind Benders

Enemy Defenses (continued)

Mind benders are small tofu-like structures that have a hallucinatory effect on the human nervous system. These devices blaze with flashing colors and spin slowly about the ball in their centers. They are extremely tough, and can only be taken out with a bomb. In many missions they are unreachable from the air. In this case a volunteer must land his plane in enemy territory and proceed on foot to plant an assassin bomb on the device, then blow it up remotely from a distance.

Aliens

Aliens are very light and like to be in constant motion. They rarely are seen on open ground and must be approached on foot most of the time. The hand gun is the weapon of choice for assassinating aliens. Upon death, chemical reactions take place that may cause the body to float up into the sky.

If an alien touches you while you are not in your plane you are captured.

In the game, aliens are inactive and undestroyable until you have approached them on foot. Then they begin to bounce about and can be shot with the hand gun. Alternatively, the pilot may re-enter his/her plane, and make effective strafing runs against them. The cannons are more effective than regular rounds.

Ground Transports

These ground vehicles look like three wheeled radio controlled cars. They are no threat to the airborne pilot; however it is a different story for the pilot on foot. These shopping carts (unofficial name) can capture an ambulatory pilot. Some of the higher quality carts are very persistent. There are two ways to avoid capture while on the ground. One is to use your handgun to destroy approaching carts. The other method is to lead them into some wreckage or other structure that will cause the cart to turn onto its side and die. Ground vehicles are notoriously stupid and do not appear to be capable of avoiding obstacles.

When a shopping cart comes near, a human on foot feels a force inexorably pulling him towards the cart. The human must escape this cart or be captured and removed from Earth.

Ground transports are exuded from triangular sites that appear to be nothing but simple triangles lying on the ground. The color indicates the quality of the cart produced. At any one time the number of carts in the area should be roughly equal to the number of sites in the area.

The shopping carts with the flashing white antennas are special elite carts. These vehicles can pass through obstacles with no damage, and can only be destroyed with guns, bombs or missiles.

Enemy Defenses (continued)

Flying Saucers
  Saucers are the primary vehicle for alien airborne transportation. They vary widely in self-defense capability. Some have AAA built in, some have missile launchers (KLA) aboard. All but the dumbest have the ability to execute evasive maneuvers. All saucers have the ability to make bombing runs against airfields. The color of the top ball on the saucer indicates its intent.

  All saucers make a peculiar warbling noise. As you damage the saucer, the pitch of the noise will deepen, and quit altogether upon destruction.

Saucers emanate from saucer sites. These sites look like large square cookie tins with an x in the bottom.

Portals
  The aliens have developed a means of instantaneous travel. Portals are small force fields that look like rectangles with the corners clipped. There are several different types of portals. Some are used by saucers to appear directly over the home airstrip. These can be used by the clever pilot as a means of getting home when his plane is not serviceable enough to fly back.

  Other kinds take you to a random place in the theater, depending on whatever the aliens used it for last. Others are placed as traps that result in a destroyed plane, or worse, instant death. Flying through one of these tiny portals can be difficult. Do not point your gunsight directly at it, especially at a low airspeed. If you do so, you will surely fly under it. Remember that to keep a constant altitude, you must keep your nose above the horizon to produce lift.

Fences
  Fences are harmless objects that appear to be a means to keep idiot shopping carts from accidentally destroying other enemy structures. They have no effect on us except that a collision with one will destroy a plane.

How to Take Off, Land, and Fly Level

The key to success in Corncob is to become a good flyer. Only when a pilot learns the abilities and limitations of his plane can he make the complex maneuvers required to make it to the rank of General.

Takeoffs

Pulling your plane up into the air is the easiest part of flying. Merely put the throttle to the maximum, and pull up on the stick when you reach over 150 mph.

However, there are some details of which the pilot should be aware. If your plane is not lined up on the runway, you will have to maneuver on the ground. To steer your plane while taxiing the joystick is useless. The joystick controls the tilt of the plane; it cannot turn the plane. Therefore, use the rudder controls to steer while on the ground. Keyboard controls for the rudder are the x and z keys, as well as the 7 and 9 keys on the keypad.

To bring the plane to a stop you must use the brakes. These are the left Shift key, the . key, or the 0 key on the keypad. The plane must be brought to a complete stop to qualify as a landing.

Once you are lined up, bring the throttle up to max. When you reach 100 mph, the tail of your plane will rise off the ground. Keep the plane on the runway by tapping the rudder keys if needed. When you surpass 150 mph, pull back gently on the stick. If you pull back too hard, the tail will swing down so quickly that it will slam back down onto the runway, thus kicking down your nose due to the bounce. This can cause a crash, so be careful.

Level Flight

When in the air the most important thing is to avoid crashing into the ground. To achieve level flight you must keep your gunsight above the horizon. How high you must keep your nose depends on your airspeed. Near the stall speed of 100 mph, the nose must be kept much higher above the horizon than when you are flying at 350 mph.

There is an altimeter and a climb rate meter to help you fly level. A warning light will flash if your altitude drops below 150 feet. When climbing, keep an eye on your airspeed. If this drops below 100 mph, then you may not have enough airspeed to fly. This makes a stall possible.

Stalls and Advanced Aerodynamics

A stall occurs when the air traveling over the wings becomes turbulent. This results in a dramatic loss of lift, and unless corrected, the plane will invariably fall to the ground. Stalling can be brought about by letting your airspeed drop below stall speed, or it can also occur as a result of an imprudent aerobatic maneuver.

How to Take Off, Land, and Fly Level (continued)

Ordinarily one thinks of the wind striking a plane from directly head on. However in real flight the wind must be coming from a direction slightly below the nose. This is required because the wind must hit the wings on the bottom surface to push the plane up to counteract the downward pull of gravity. The variation of the wind direction from the directly head on direction is called the angle of attack. The greater the angle of attack is, the more the wind direction appears to be coming up from below the nose.

The greater the angle of attack, the more lift your plane feels. When you pull back hard on the stick, you are tilting up the nose of your plane while tilting the tail downward. This act changes the apparent direction of the wind. Before you pulled back on the stick the wind was coming almost head on. After tilting your plane's nose, the apparent wind is coming from beneath the plane. The wind hitting the bottom of wings is what starts lifting the plane (accelerating the plane in the upward direction). As the nose lifts, you are increasing the angle of attack, thereby increasing the lift, and the plane will begin to rise.

Once the plane has accelerated in the upward direction, then the apparent wind angle lessens. This is because the upward speed of the plane has changed the apparent wind direction. (Sailors are familiar with this effect; the apparent wind direction depends on what direction and how fast their boat is moving.)

Stalls occur when the angle of attack is too great. At this point the air no longer travels smoothly over the wing. This determines the lowest flying speed of a plane. The lower the airspeed of the plane, the greater the angle of attack must be to maintain level flight. If the airspeed is too low the plane begins to lose altitude, thus increasing the angle of attack (just as the upward motion of the plane decreases the angle of attack). If the pilot then raises the joystick in an attempt to stop losing altitude, that angle of attack may go beyond the stall angle, thus causing the stall.

In a real plane a pilot can use his senses to judge when he is close to the stall speed, but in Corncob there are just the instruments. In a real plane the pilot knows how much force he can put safely on the stick.

However, in a computer simulation, the joystick tension does not reflect the flight situation. Therefore, Corncob has in it an automatic stall preventer that limits the stick travel to the maximum attainable without causing a stall. This does not mean that it is impossible to stall. The stall preventer simply limits the control stick from causing a stall which

How to Take Off, Land, and Fly Level (continued)

would be the result of pulling back too hard.

The most common cause of a stall is simply that the airspeed dropped too low to sustain flight. Another common cause is excessive sideslipping. The aerodynamic model that Corncob uses includes sideways motion of the plane through the air. When the rudder is used it kicks the nose of the plane to the right or left, without necessarily changing the direction of motion of the plane. This changes the apparent wind direction to be either to the left or right of head on. This sideslipping can also result from turning the plane on its side, and letting gravity pull the plane downward. (You obviously cannot fly level with the plane on its side, since then the wings' lift is not pulling against gravity.)

If the plane is in a fast sideslip, and then the plane is quickly righted, and the stick is pulled, the momentum of the plane can result in the effects of left-right wind angle being added to the up-down angle of attack, thereby pushing your angle of attack over the stall angle.

Also, a damaged plane can be much more likely to stall. Wing damage will lower the lift of the wings, and control damage can make the stick jitter beyond what the automatic stall preventer can handle. Of course engine damage will lower airspeed, and rudder damage will increase level flight sideslip.

How to Get Out of a Stall

To get out of stall means getting your forward airspeed up. Since your direction of motion is invariably down when stalled, the best idea is to get the nose pointed down. Of course your throttle should be place to maximum also.

Getting out of a stall can depend on interpreting the airspeed direction indicator. The red ball tells which direction the apparent wind is coming from. In level flight it should be riding not in the center, but below center. When you yank back on the stick you will see the red ball drop as the angle of attack increases and the opposite when you slam the stick forward.

When you are slipping sideways the ball will go to the left or right. Try hitting the rudder keys to watch as the apparent wind direction moves to the left or right, then snaps back. It snaps back because all stable planes are balanced to have the plane automatically turn into the sidewinds. How fast it snaps back depends on your airspeed.

So now that we understand the red ball, here is how it can help you. When the red ball is somewhere near the center you are OK. When stalled the wind direction can be so far off center that it disappears from the display altogether. To become unstalled involves getting that red ball back on the screen. If you were doing a hairy maneuver at the time, try just reversing the stick to try to undo whatever aerodynamic motion you

How to Take Off, Land, and Fly Level (continued)

did last. If you are below 1,000 ft, consider ejecting. It takes time and altitude to recover from a stall.

Note that when the stall is so bad that your airspeed is nil, then you have lost the ability to control the plane with the control surfaces. In fact, it is even possible for your plane to fly backwards with the controls operating in a reversed fashion. Even backwards airspeed is better than zero. If you are falling backwards, pull the stick back to push the nose down. Once the nose is down, hold her there until you get some forward airspeed.

Landing

This is one of the hardest parts of learning to fly. This requires much patience, but in time you will be able to swoop down from the vertical direction and pop the plane down in seconds.

The biggest problem for beginners is to get lined up properly. One point to remember is that you must not fly toward the middle of the runway. Fly towards the close end of the runway, or a point on an imaginary line running the length of the runway and beyond. Once you are over that imaginary line, then turn your nose to point towards the end of the runway. Getting lined up is especially difficult in Corncob, due to fact that the plane may be slipping sideways.

When you get close enough to the runway where it starts to switch from being drawn as lines, to being drawn with filled black rectangles, engage the flaps, and drop your throttle to zero. (But if your airspeed drops below 100 mph you may need to bump it up again.) At this point you should be no higher that 1,000 ft.

In Corncob your plane has automatic landing gear that pops down as required. (It's actually a force field gear using stolen technology.) The pilot does not need to explicitly engage the gear.

Now let the plane drop, shedding that altitude before you reach the end of the runway. When you get to 100 or 200 feet, level the plane out for the final drop to the runway. If you are not lined up well, be ready to punch the throttle and take off for another pass. This is very important, because with your throttle dropped, once you have leveled out, the plane begins to decelerate. As the airspeed drops below 100 mph your plane becomes very unresponsive to the stick.

If you are not perfectly lined up, but are low and over the runway, then drop and steer with the rudder once your wheels touch. The gear of your plane is quite tough, and on the runway surface you can have your rate of climb meter almost pegged negative and still survive. Landing on the grass can be done, but requires a much more sensitive touch down.

How to Take Off, Land, and Fly Level (continued)

It is a good idea to be holding down the brake key throughout the whole landing process. The left Shift key is situated conveniently for the left hand to operate the brakes and rudder controls. For those that use the left hand on the joystick, use the keypad rudder and brake controls.

Once you have solidly touched down you may taxi about using your rudder. Be careful to keep your speed down while taxiing (especially if your flaps are down) because it may cause you to lift into the air inadvertently, and then cause a crashlanding. Remember that on the ground, the stick is useless, and you must steer with the rudder keys. Pulling the stick over on the ground may cause your plane to flip and cause death.

There are two different types of crashes. One is a fatal wreck and the other type is a crashlanding. A crashlanding is where your plane is brought down too hard, and the landing gear fails. The plane then spins to a stop on the ground. When it stops, your plane is dead and will not fly again until repaired. If you are close to your airstrip the repair crew can fix it. (Do this by entering the control tower through the door on foot.)

The other type of crash is the fatal crash. This happens if you slam into the ground too hard, or if your plane is so tilted during landing that your wing tip hits.

Midair Collisions

There are many types of objects that will destroy your plane if you collide with them. When this happens your plane bounces and the damage indicators immediately max out. The screen border turns dark red, and the eject light begins to flash. At this point you should bail out, since it is likely that you will die if you try to land the plane. Don't bother trying to make it back to base. (Although landing such a plane has been done on occasion, it most often proves to be a fatal mistake.)

Some of the objects that will destroy your plane in a collision are: saucers, fuel dumps, TRFRUs, and generator orbs.

How to Turn Your Plane

A combat aircraft is a very heavy object. (A Corsair weighs 14,000 pounds.) Changing the course of such a massive object takes a huge amount of force. Negotiating a turn requires an aircraft to be pushed laterally with a large force. Although one might think that the rudder pushes the plane through a turn, this is not true. The rudder and the other control surfaces only rotate the plane; to change the plane's flight path requires using the force of lift from the wings.

An airplane turns very differently than an automobile. A non-skidding car will instantly travel in the direction it is pointed. A plane however can easily slide sideways with little resistance. Think of the plane as a car with greased metal tires. When the driver turns the nose of the car it will simply slip sideways. But if the driver could find a nice banked curve, he could use the sideways force from the bank to provide the sideways force needed to push his car around in a turn, thus turning his car quickly without needing good traction.

An airplane turns using this method. The pilot banks the aircraft, thus using the tendency of the plane to sideslip to the downward side to exactly counteract the sideslip caused by turning the plane's nose. So the sideways force needed actually to make the flight path of the plane follow a turn is provided by making the huge lifting force of the wings pull the plane sideways.

How to Turn Your Plane in Corncob

Corncob's advanced flight model makes turning different than on many commercial flight games that have simpler models. With a little practice, however, you will find the plane is just as maneuverable.

To turn your plane simply move the stick left or right. The plane will begin to bank. Don't be too shy, give the plane a nice 45 degree bank. Now that the wings' lift is no longer pulling straight up, you will lose some lift, and your nose may drop. Pull back on the stick to keep the nose above the horizon. You will notice that as you pull back on the stick, your nose will be pulled into the turn. To make a typical combat turn, place the plane on its side, so that the horizon forms a vertical line. Then pull all the way back on the stick, and hold it there until you have the compass reading what you desire. Then simply flatten your plane back to the level. Be careful not to let the nose drop during the turn. You can always bring it up by unbanking the plane a little, and the nose will lift since you are holding the control stick back. This method of turning is actually easier than a slow coordinated turn. The price for the fast turn is airspeed.

The fastest turns are done with the flaps engaged, since this greatly increases the lift from the wings. Be careful not to let your airspeed

How to Turn Your Plane (continued)

drop too much. Flaps increase drag also.

  Note that the amount of time required for the turn depends very much upon your airspeed. (Just like in an automobile, when moving fast you cannot make tight turns.)

Dealing With Air Density and Gravity Variations

In the Corncob Other Worlds Campaign you will be flying missions on a wide variety of planet types with different sizes and atmospheres. Here are some of the simpler ways your plane is affected by these variations.

Thin Air

The plane will maneuver more sluggishly in thin air. Your nose will have to be kept much higher above the horizon to fly level, and stalling is much easier. Flaps are a must, as well as plenty of airspeed. One good thing about thin air is that the air drag is much lower, thus enabling higher airspeeds and acceleration.

Thick Air

Thick air improves maneuverability and increases air drag. It is much harder to stall. Thick atmosphere is in some ways equivalent to flying with the flaps engaged. Maximum airspeed is lower in thick air. Your stall speed is lower, thus making landings easier.

Weak Gravity

Weak gravity raises maximum airspeed of level flight, and the nose is kept closer to the horizon. Certainly climbing is much easier, and in some cases, continuous vertical flight is possible. Caution must be exercised, however, as the unwary pilot may find himself in orbit by mistake.

The good news is that in a stall situation the pilot will have more time
to get the plane under control before hitting the ground.

Heavy Gravity

Strong gravity decreases maximum airspeed of level flight, and the nose
is kept farther above the horizon. Certainly climbing is much harder. Keep in mind that your stall speed will be much higher, and that could make landings a little more difficult.

The airspeed picked up in a dive will be more significant, and the time
you have to get out of a stall before plummeting to the ground will be much reduced.

How to Use Your Missiles, Bombs, and Guns

Aiming

Your main goal in Corncob is to destroy alien forces. You accomplish this by using your guns, missiles, and bombs against enemy targets. In addition, you must sometimes use your offensive weapons for self-defense.
(For example, your guns can explode approaching AAA shells, and enemy missiles).

The most obviously important requirement for successful gunnery is aiming. Your gunsight shows where the bullets will strike a non-moving target. There is much technique involved with holding the gunsight on target, since your plane has to follow the rules of aerodynamics. This technique will have to be developed over time and cannot be learned by any other way but experience. However, don't be afraid to use the rudder. The rudder is an excellent way of turning your nose fast for a quick shot to the side. An experienced pilot who uses the rudder can destroy twice as many targets as the pilot who uses the stick alone.

As far as shooting moving targets, don't forget to lead your target properly. By the time your bullets reach the target, most likely the target will be somewhere else. When attacking a KLA for example, try to judge where it will be when the bullets get there, and fire ahead of it.

When shooting at ground targets the most important thing to remember is to watch your altitude. Remember that you must keep your nose above the horizon to maintain altitude, so that means you can only shoot at the ground for a limited time before breaking off to regain altitude.

Both your guns and missiles can be used continuously, but after an initial burst, the firing rate will drop to a slow plodding beat. If you want a better chance of taking out something quickly, wait until you are on target, and release a volley.

An important consideration when attacking fixed targets is rate of closure. To destroy something in the least amount of time, you would like to make as few passes over it as possible. If you come screaming in over the target at 350 mph, then chances are you will get in only a few shots before you must pull up and around for another pass. That's why it is sometimes a good idea to cut the throttle, and throw down the flaps when engaging a ground target. With an airspeed of 150 mph you have twice the amount of time to pound the target. Many times this gives you enough time to take out the mission objective in one pass. If you do this, however, be sure keep an eye on your airspeed. As soon as you level out or start to climb, your airspeed will quickly fall below stall speed unless you remember to slap the throttle back up.

Guns vs. Missiles

When should you use guns, and when should you use missiles? Guns are fast, delivering a stream of destruction. Missiles take time to accelerate and deliver isolated punches of firepower.

How to Use Your Missiles, Bombs, and Guns (continued)

Although missiles have more punch, they have a tendency to blow up in your face if used at the wrong time. Missiles explode when any object is within their proximity. This makes them less sensitive to aiming errors but also dangerous to use. Never use missiles when close to a forcefield.

The fields will often blow your missiles up us soon as they leave their launch tubes, thus causing brutal damage to your plane. This can happen even though your plane is well away from the forcefields. Never use missiles alone to engage AAA sites. The shells from the AAA will be flying so near you that the missiles may explode directly in front of you as they encounter AAA shells. The combination of guns and missiles works well, though, since the guns explode the shells long before they come near your plane. Be sure to open up with guns first, then start the missiles.

Never use missiles when you are too close to the target. The explosion will damage your plane.

Your guns are almost always safe to use. Although the bullets may also explode when they contact forcefields, they are safer to use. Most explosions caused by gunfire are safe to fly through, so it is the weapon of choice for close engagements. They are particularly good for taking out KLAs, since they are easier to aim when leading the target and send a stream of firepower, rather than discrete charges. If a KLA is charging down your throat, use guns, not missiles.

Most of the time a combination works well. On tough targets, use the guns first, and when you start getting hits on the target, send a volley of missiles.

Bombs

Although your plane can create unlimited bombs, they regenerate at a slow pace. Your bomb display shows how many you have available to you. The maximum number of bombs available at any time is three.

These bombs cause an initial impact, and then release clusters of bomblets that cause additional damage. As you watch your bomb explode in the rear view, you can see the yellow bomblets take to the air, then fall back to earth. So although you need a direct hit to cause serious damage, the bomblets can damage neighboring targets. If the bomb detonates near the ground, a crater is formed. This crater is very hot, and will continue to damage targets in contact with the crater at a constant rate until the crater disappears.

The bombs are very light for their volume and decelerate quickly once they leave the plane. There is a lot of technique for making successful bombing runs. In Corncob, most of the bombing is divebombing, since there is no bomb sight for high altitude bombing. Most of the time your plane

How to Use Your Missiles, Bombs, and Guns (continued)

is at about 200-600 feet, and just as your target disappears from your view as you pass over it, release your bombs. Then pull up fast to get away from the blast and to watch the bombs hit in the rear view. The craters will tell how close you were to the target.

  Note that exactly when to release depends heavily on your airspeed and altitude, as well as your direction of flight during the drop. This will
simply have to be learned via trial and error.

  As with missiles, bombs are proximity devices. This means you should be
careful in the presence of forcefields or AAA shells. Bombing high-quality AAA sites is not always such a good idea, since the AAA shells can explode the bomb in your belly, and also because you are a perfect target as you fly away.

  Be careful when dropping bombs at low altitude, especially when your airspeed is low. The bomblets reach hundreds of feet into the air.

  Bombs have been successfully used against KLAs that are right on the tail, and against saucers, but this sort of bombing is very tricky. Bombing shopping carts is great sport, because you can take advantage of the fact that some kinds of them follow your plane fairly well.

  Bombs can be dropped from any position, even while inverted. The bomb delivery mechanism is very flexible.

Defensive Flying and Bringing a Damaged Plane Home

Defensive Flying & Making it Home With a
Shot-up Plane Bringing your plane back to base is necessary to earn a good score. Although any clown can make effective suicide runs into enemy
targets, it takes a seasoned pilot to weather the enemy storm, and still make it back to base.

The two most important air threats are deathballs and AAA sites. To avoid being shot out of the sky by AAA requires some facts about AAA. In
Corncob the aliens' AAA shells glide slowly up from the AAA sites, without
gravity effects. The shells can be seen as brown dot or spheres fly upwards towards you. Coming near anything will set off the AAA shells, and bullets will certainly do the job. If you are attacking a lone AAA, or a close bunch of AAA, your best defense is a good offense. Blowing away those approaching shells with gunfire is the most effective defense.
This technique is limited in the presence of KLAs however, since you cannot stay flying in a straight line when deathballs are charging up on your tail. This method will also fail for widely dispersed AAA, since the
shells coming from the side will be pounding you as you stay locked on another installation.

These shells are equally capable of destroying alien targets as well. As result of that fact, ground based AAA will only fire so low. Anything
on the ground should be safe from AAA. It is possible to fly under their range of fire. This means flying under 150 feet (or lower if you are close to the site). This method is also limited in the presence of enemy
KLAs, since you will need room to do evasive maneuvers.

AAA sites have the same problem as any other gunner does when the target
is moving. This is the problem of leading the motion of the target, so that you are actually firing at where the target will be, not where it is
now. The aliens have 8 different qualities of AAA. The lower 4 qualities
do not have to ability to lead targets. Therefore, if you can always be flying with the AAA on your side, you will not be hit if you keep your airspeed up. The higher quality AAA can lead, so to avoid them you must change your flight path often in such a way as to make them anticipate incorrectly. The target-leading AAA assume you will fly in a straight line at your present speed. If the AAA is on your side, and you fly up and down, you may lessen the hits. In the case of high-quality AAA, however, the only techniques that work well are the first two.

KLAs are not hard to evade no matter what their quality is, provided there is only one of them in the area. Deathballs are enemy missiles that
always try to fly directly at you. Since they do not lead their target, they will invariably end up chasing you from behind. When a KLA has a lock on you it will flash. In addition to flashing, it will also turn on
its maximum thrust, and try to come screaming in to finish in a blaze of

Defensive Flying and Bringing a Damaged Plane Home (continued)

glory against your hapless plane.

Although this death charge can be very effective, it can also be used as a defense for you. If you wait until the last second before impact, and then pull up as hard as you can, the KLA cannot turn fast enough to follow due to its high speed. The spitball will then pass under you, enabling you to snap your nose down and blow it up with your guns. If there are a number of foes forming a train behind you, this technique will lead to disaster. Although you will have evaded the first in the train, his friends will surely eat you alive when you punch your nose back down.

The solution is to pull up, but continue the loop in hopes of shaking the whole train. In a sky full of missiles, this works for a while, but you must eliminate their sites, because eventually you will become toast unless you can dwindle their numbers. Knife fighting like this with deathballs is very dangerous in the presence of AAA sites. Your airspeed will drop quickly with this much maneuvering, and a fumbling plane in the air is an easy low-deflection target for even the stupidest AAA.

When your plane has been trashed miles from the nearest friendly airstrip, don't despair, plenty of pilots have made back to base with planes that were just about as flyable as a big stone. (Landing a plane that has collided with a saucer has even been done, but landing a plane that has more than 100 hits depends a lot on luck.)

The two most important factors when you are trying to make it home are airspeed and altitude. The worst possible situation is to be very far from your base flying 100 mph at 500 feet. Usually there will also be more KLAs in the area, and evading them with a damaged plane can be impossible.

But even in the worst situation all hope is not lost. The clever designers of your modern plane have installed a rocket motor, which can give you the altitude and airspeed you need to make it back. Look at your turbo boost gauge. The needle should be green, indicating the rocket is charged and ready. Hit the i button and start climbing.

If your flaps are engaged, tap them off, but only after engaging the rocket motor, since it is possible you could stall just by pulling back the flaps.

Your rocket motor will burn for 15 seconds. The turbo boost gauge needle will be red during the boost and go to blue when the charge is spent. It will take about 120 seconds to recharge the booster.

Now concentrate on keeping the slowest descent possible while maintaining airspeed. If KLAs are screaming up on your tail, it may be best to let them hit you, if you think you will certainly stall. Don't forget to try dropping a bomb when they get very close. Many times this will take them out without having to maneuver.

Defensive Flying and Bringing a Damaged Plane Home (continued)

When the turbo boost gauge needle goes back to green, you may want to go
for another rocket thrust. Never use the rocket thrust while on the
landing approach, since you cannot stop it once you have started a burn.
Unless of course you wish to abort the landing attempt, and need airspeed
quickly to avoid a stall.

Running Around - Welcome the Assassin

Part of your job as a pilot involves special forces missions. You may have to land in enemy territory and continue on foot. You carry a pistol with twenty rounds and one assassin charge; boarding a plane resupplies you, including a crashed plane.

Move with WASD and look with the mouse. Space jumps immediately; you can steer in the air. Q opens the parachute, even during a jump, and Left Ctrl
lets you drop through a platform. Horizontal force fields, fuel dumps, platforms and control towers may support your weight. Watch your landing:
a long fall can still kill you.

Plant a charge with Semicolon and move to safety. B arms the charge and lets it fall; contact then detonates it. A charge can be planted in midair.
Mindbenders are vulnerable only to bombs, so some missions require you to
plant a charge inside one and retreat before arming it.

Shopping carts and aliens can capture you. Use your pistol, jump to safety, or lure a cart into wreckage. V calls a rescue van. Caps Lock switches remote aircraft control; it does not move you back into the cockpit. Walk near a parked plane to board it automatically.

Scoring and Promotion Rules

Scoring
  Your score for a mission is determined by a formula outlined below. Your raw score (which depends on how much stuff you killed) is multiplied
by 3 factors. These three factors depend on what happened during your mission. For example say your raw score for a mission was 1,000. Further
suppose that you returned from your mission, and landed your plane safely
at an airstrip. Then your final mission score would be:

5 * 5 * 1 * 1,000 = 25,000

But say you ejected in combat and had to be rescued by the rescue van.

4 * 1 * + * 1,000 = 2,000

  Here you can see that bringing your plane home increased your score by 10 times.

  Now let's say you brought your plane back but didn't have the guts to try and land it.

5 * 1 * 1 * 1,000 = 5,000

  So you can see it is worth it to learn how to land the plane. It may be
impossible to get a Stellar Cluster without landing your plane.

  If you make multiple flights during your mission, the multiplier reflects the worst thing you did. For example, if you crashland a plane,
but then take another into the air (without quitting the mission) and do a
great job, and even land beautifully, your multiplier will reflect the fact that you crashlanded. The fact that you also landed will not affect
your score.

  Therefore you can see that flying against multiple mission objectives may raise up your score, but you are jeopardizing the points you have already earned by risking a crashlanding or worse. Of course you'll never
top the high score list without taking some chances.

  Technical Score Calculation
  The score is a product of four factors.

  score = pmsf * msf * nplf * mobf

  pmsf is Present Mission Status Factor (KIA, MIA, HOME, RESCUE).

  msf is Mission Status Factor (unhome-crshlnd,eject inflt,crsh-home,land-home).

  nplf is number of Planes Lost Factor, which = 1/(number of planes lost +

Scoring and Promotion Rules (continued)

1).

  mobf is Mission Objective Factor (number of kills).

Promotions
  One of the most exciting aspects of Corncob is the possibility of promotions and medals. Once you have started a character, he can progress up the ranks, with the ultimate goal of achieving general. Making general with a pure character (i.e. one that has not been killed and resurrected) is the ultimate challenge in the game, and will rank you at the absolute top in bragging rights.

  Promotions in Corncob are based on the number of completed mission objectives (MO), and go as follows:
  2nd Lieutenant: 0 MO. (You start at this rank.)
  1st Lieutenant: 1 MO.
  Captain: 8 MO.
  Major: 20 MO.
  Lt. Colonel: 20 MO in at least 2 different Theaters-of-Operation.
  Colonel: 20 MO in at least 3 different Theaters-of-Operation.
  General: ALL MO in ALL required Theaters-of-Operation, plus successful completion
      of the Secret-Mission.

  Note: Required Theaters-of-Operation are marked by a Yellow Block in the list at the upper-right hand side of the Theaters-of-Operations menu.

Medals
  Medals are based on a variety of criteria. Medals are listed in the pilot summary shown in the roster, and T-of-O menus when a pilot or theater is viewed. In this case, the Medal is listed as a short abbreviation, such as BS for Bronze-Star. If the pilot has more than one of the medal, the abbreviation is followed by a suffix indicating how many such awards the pilot has, such as BS:2 for 2 Bronze-Stars.

  A complete list of all the medals and their abbreviations follows:

  Purple Heart: PH Awarded for ending a mission with more than 27 units of damage to
      your plane, without losing your plane, and without crashing unless at home.

  Service Award: SA Awarded for surviving 10 or more missions.

  Bronze Star: BS Awarded for achieving a score of 10,000-19,999 points in a mission.

  Silver Star: SS Awarded for achieving a score of 20,000-49,999 points in a mission.

  Gold Star: GS Awarded for achieving a score of 50,000-99,999 points

Scoring and Promotion Rules (continued)

in
a mission.

  Stellar Cluster: SC   Awarded for achieving a score of 100,000 points
or
more in a
      mission. (This is really exceptional!)

  Distinguished Service Cross: DSC   Awarded for accomplishing 5 or more
mission
       objectives in a single mission.

  Theater Combat Medal: TCM   Awarded for completing ALL mission
objectives in a
       single Theater-of-Operation.

  Distinguished Flying Cross: DFC   Awarded for completing ALL mission
objectives in a
     single Theater-of-Operation without losing a plane. This takes a
real hot pilot, as well as
       some consistent luck.

  Combat Master: CM Awarded for completing ALL mission objectives in
ALL
of the
       required theaters. This is a prerequisite for making General.

Commonly Asked Questions

> Sometimes I can't get back into my plane. I just go right through it.

  There are two possible reasons for this. One is that you must walk at least a couple plane lengths away before attempting to reenter. This is to avoid popping back into your plane before you had a chance to walk away.

  The DOS version could also lock planes after walking far away; that renderer-dependent restriction does not apply to Redux. Historically, If
you stray so far from your plane that it has shrunk to a point, then the plane will engage the automatic security system, and lock itself up tight.
 In none of the missions is it imperative to walk anywhere near this far from your plane.

> Sometimes when I eject or crashland, there is a continuous tone that never fades.

  Under certain conditions, the stall indicator sound can get stuck on. Just tap the s key twice to reset the sound system.

> Sometimes the game refuses to take me to a mission via F2, even though I
select it properly, and my plane has no damage.

  Each mission which can be teleported to has a line in the intelligence briefing that starts with Sxyth. When this line lights up, then you are ready to teleport to that mission. Sometimes when mission briefings are short, two Sxyth lines from different missions can be on the screen at
the same time. In this case, the top Sxyth line will light up, and it is
this mission which you be teleported to. If you desire the mission that is lower down on the screen, you must move down with the arrow keys until
the mission above is scrolled off the screen.

  By the same token, if the description of the mission is very long, and you have scrolled the Sxyth line off the screen while reading the description, the teleport will not work. Teleportation will always bring
you to the mission with the lit-up Sxyth only.

> When first lifting off the ground, why does the nose lift for a second,
and then seem to knock back down?

  If you pull back too hard on the stick while lifting off, your tail wheel will be slammed back down onto the pavement, and when it bounces upward, your nose appears to bounce downward. It is possible to crash and
burn as a result of this, so pull back on the stick gently.

> Why does the stall light go on sometimes when I'm taxiing on the runway?

Commonly Asked Questions (continued)

The stall detector is looking at the airflow over the wings. It is calibrated to detect stalls when the plane is flying. When on the ground, the weight of the plane is held by the landing gear, thus making the stall detector give incorrect readings. Just ignore it.

> Why doesn't the plane steer worth a bean when I'm on the runway?

Don't try to use the joystick to steer a grounded plane. The control stick left/right controls mostly only the roll (left/right tilt) of the plane. You must use the rudder controls to steer on the ground. These are the x and z keys, or the 7 and 9 keys on the keypad.

> Why don't the brakes appear to work?

In Corncob, the brakes do NOT toggle on/off. To stop you must hold down the brake key until the plane comes to a complete stop. The default binding is Period or keypad 0.

> How does the T key accelerate time?

Hold T to run the whole simulation ten times faster, on foot or in flight. Enemies, projectiles and production advance at the same rate; they no longer sleep while you travel. Release T for normal speed.

> Why does using the rudder in flight just make me stall?

Unlike other flight sims, Corncob's flight model includes the capability of your plane to slide sideways through the air. The rudder causes your nose to flip to the side, without necessarily changing the direction of motion of the plane. Using too much rudder (by holding the key down too long), will cause the plane to turn to the point where it is hurling sideways through the air. Note that as soon as you release the rudder, the sidewind makes the nose tend to flip back into the sidewind, thus restoring the original orientation of the plane. The rudder is useful for flipping the nose a little to the side for a quick shot at a side target without the need of negotiating the slow process of turning the plane.

> How do I know when to end the mission?

A large part of Corncob is bringing your plane back to the base. Therefore you must bring your plane to a friendly airstrip, or else when you go to fly again, you will have one less plane in that theater.

To exit without losing a plane, choose End current sortie from the pause menu when you are flying low over your airfield (with the radar tower flashing white), or else land on the air strip. Landing on the airstrip will give you a much higher score multiplier, so it is worth an attempt if you think you can do it without crashing.

To find an airfield, use the map that is brought up with the m key.

Commonly Asked Questions (continued)

> How do I zoom down closer when in map mode?

When in map mode, the KEYPAD 9 key will zoom down, and the KEYPAD 3 key will zoom up. Use the joystick or keypad arrow keys to move around. To make the map jump back to your present position, hit the KEYPAD 7 key.

> Why do I die every time I eject, regardless of altitude?

Ejecting successfully from a flying plane requires two keys: E to eject and Q to pull your ripcord. If you don't pull your ripcord, you will die when you strike the ground. If you bail out at 20,000 feet, wait until you get closer to the ground before pulling your ripcord; otherwise you will be stuck in the air for a long time.

> Why are some generator orbs seemingly indestructible?

Some orbs have been designed to be invincible to normal weapons, and the only effective way to take them out is via suiciding the plane into them. Make sure you bail out before contact, or else you will be killed in the ensuing explosion. If you must bail a good way away from the target, hit caps lock to control your plane remotely. The indestructible orbs are a light shade of blue/green.

> Why does the complete light refuse to go on even though I know the mission is completed?

The mission objectives have to be given some geometric scope via the mission builder when the mission is created. Almost all mission complete sensing is done via the generator orbs. If you have completed a mission, but the complete light doesn't go, make a flyby of the mission's generator orb (or it's remains), and many times the complete light will belatedly light up. Note that if your complete light never lights, you will not get credit for accomplishing an objective.

> Sometimes my plane goes into a mode in which my keypad arrow keys stop working, and something weird happens.

Numlock toggles the keypad arrow keys between views and control of the plane. Note that the numlock light on the keyboard will not change. If you would like to use the views without changing the mode of the arrow keys, hold down the left ctrl key to temporarily change the function of the arrow keys to view, the normal function of the keys returns when you release the left ctrl key.

> I can't find any way to put down the landing gear!

The plane has landing gear that comes down automatically. From a game

standpoint, it seemed unnecessary to have something that doesn't add to the fun of the game and can cause pilots to be killed. We all get punished in life enough for being absent-minded; why be punished when playing a computer game?

> Why can this Corsair do a loop directly after take off?

Because the number one priority in this game is to have fun. That meant increasing the thrust of the engine by a factor of four. For those sim buffs who will criticize that, how many flight games keep the realistic figure of 10-15 minutes just to climb to 10,000 feet? Who wants to spend 1 hour of his life just sitting there watching your computer plane climb?

> I have completed all the mission objectives in a theater, but the computer insists there is still one more.

Many times you really have forgotten a mission. Use the map to make sure all the targets are destroyed. Remember that many missions don't show up unless you have zoomed down for a close up with the Pgup key.

Most things should show up when one grid is about the size of the screen.

Another possibility is that a mission-objective type saucer has wandered away during the melee, and is now somewhere unexpected, perhaps 10's of miles away from the original mission site. Track it down with the map feature. The new tower screen from the fly mission menu should tell you to which airstrip the saucer is closest.

The DOS Other Worlds manual also describes an F6 in-flight object list. In Redux, F6 selects the cockpit or first-person view instead. For an object list, use the map, tower briefings and the live object survey (I on the starting-airfield screen).

Many times a secondary mission objective is mentioned in the mission briefing. This secondary mission also must be completed before a theater is complete. Often this is a generator orb. All orbs with a black base must be destroyed to finish a theater, whether the mission information directs you to do this or not.

General strategy and flying hints

Attack Priorities:

When attacking an installation protected by both KLA's and AAA, first attack the high-quality AAA. These defenses will chew you up the most quickly while undergoing evasive maneuvers. Then go for the top-notch KLA sites. Once these threats have been dealt with, proceed with the rest of the defenses at your leisure.

Enemy activity at a distance (Redux):

Enemies, projectiles and production sites continue to update when you leave their area. Travelling away no longer freezes their timers or repairs their damage. Weapon ranges and local production caps still apply; draw distance does not change these rules.

Use the half barrel roll to ease blackouts quickly:

When a high-g turn blacks you out, the quickest way to get that blood back into your brain is to do a quick high-negative-g turn. Simply easy up on the stick, and flip your plane around upside down, and push forward
on the stick. If you managed to flip the plane correctly, you should still be turning in the desired direction, and your vision should return very quickly.

Use dumpy planes to do high-risk platform missions:

When you have to land on a suspended airstrip and proceed on foot to a difficult assignment, it is not unusual to fall off a platform, and to leave your plane abandoned up on the airstrip. You may want to ferry your late model planes away from one of the airstrips in the theater.
Then when you fly again from the planeless airstrip, the program will ask
you if you want to pull a dumpy plane from storage. Take the dumpy plane
to the suspended mission, so you won't lose a good plane if you fall.

Avoid bounce-back on suicide missions:

Some missions involve suiciding your plane into a generator orb. Be aware that your plane will bounce backward upon impact. It is not uncommon
for the plane to bounce back toward you so that you find yourself back in the cockpit after collision, hurtling toward the ground in a ball of flaming wreckage.
Be ready to eject again in case this occurs. Take care to time your ejection correctly so this doesn't happen.
Note also, the hint given above about using dumpy planes for dangerous mission applies to suicide missions also.

Flying low over craters (Redux):

Bomb explosions and their debris remain dangerous. The lingering crater cannot hurt you or your aircraft, so it no longer creates an invisible obstacle above the ground. Its blast damage against enemy structures is retained.

General strategy and flying hints (continued)

Use rudder to reduce airspeed on runway:

  Sometimes when landing on suspended runways it is imperative to come to
a quick stop. Hold down the brake key, and simultaneously work the rudder
back and forth. The swishing of the tail will help you slow down, as
well as making your path longer to the end of the runway. Take care not
to direct the plane off the edge.

Use barrel roll to reduce airspeed on approach:

  When making a high-speed landing approach, you sometimes want to drop
your excess speed. Dropping the flaps, and cutting back on the throttle
many times isn't a big enough effect, especially if your angle of descent
is too steep. In this case, try pulling the stick all the way back, and
fully to the side. The plane will travel in a barrel roll and end up in
approximately in the same flight path, but with the airspeed down
significantly.

Techniques for escaping with your plane intact

Use teleportation to get out of trouble:

  This is a bit of a cheat, since it is an unrealistic way out of impending
doom. The purists will want to not do this, since it twists a convenience
feature of the game into an irrational combat advantage.
  If you are in deep doo-doo and are worried that you will not get out
of this situation alive, simply hit F2, and select a new mission. If you
are somewhat damaged, you may want to choose a mission which is close
to your airstrip. Also, you might want to choose a mission which you have
already completed.
  Note that to keep this feature from being abused, there is a rule that
you cannot teleport if your plane has sustained serious damage.

Use autopilot when heavily damaged:

  When your plane is a flying scrap, and you are having difficulty
keeping your plane in the air due to excessive control damage, try turning
on the autopilot. This should help you keep the plane reasonably level.
The autopilot will react faster than you, and can make the difference needed
to make it home safely.
  Beware that the autopilot will interfere with attempts at evasive maneuvers.
If you are going to try to dodge a KLA, make sure you disengage the autopilot
first.

Use the emergency rocket booster to full advantage:

General strategy and flying hints (continued)

When you are surrounded by KLA's, and you have taken too many hits to risk continuing, hit 'i' to ignite the rocket booster. If you are at low altitude, pull the nose up to vertical to get the maximum altitude gain from the rocket. However, be sure to push the nose back to the horizontal BEFORE the booster runs out. This is because you will really need the airspeed in addition to the altitude.

If you already have good altitude (for instance if you are in a high altitude mission), then use the rocket booster for pure speed. Disengage the flaps to maximize airspeed, put yourself in level flight, and let her rip. You can outrun almost any of the KLA's with the booster.

Note that it is generally a good idea to get the plane headed towards your airstrip before engaging the rocket. Tight turns with the booster running will hurt your gained airspeed by quite a large amount.

Hints for defeating KLA and AAA

Blast distant airborne KLA's:

When approaching a mission from afar, many times there will be KLA's already airborne and approaching from the front. Frequently, these pre-launched KLA's will be of higher quality than any of the nearby KLA sites. Therefore, it is usually a good idea to blow these approachers out of the sky with missiles and guns. Since they are advancing upon you head-on, it is an easy zero-deflection shot.

When NOT to shoot KLA's:

Keep in mind that KLA sites will only produce KLA's when the number of active KLA's presently flying is less than the number of sites. Therefore if you shoot them, the sites will make more. The key here is to only shoot the high-quality deathballs. There is a good chance that the next site to create a replacement will be a lower quality site. Thus by shooting the high quality KLA, you have traded it for a low quality model.

Of course, for the same reason, you should NEVER shoot a poor quality KLA if there is a variety of sites around. A high quality site may make its replacement.

Remember that the lower quality models pose almost no threat to the seasoned pilot.

Shoot at KLA sites at every chance:

When a KLA site takes some damage, it is prevented from production for a period of time. Therefore, even if you haven't enough time to get in enough hits to destroy the site, even one hit with a bullet is enough to keep it out of commission for tens of seconds.

When you see a clump of KLA sites, always spray a hail of bullets and missiles over them just to stunt their output for a while.

KLA production and time (Redux):

Production follows fixed simulation time, rather than displayed frames.

General strategy and flying hints (continued)

Low frame rates or different rendering resolutions do not slow a site's countdown. Holding T advances everything ten times faster, including KLA production and enemies, so it speeds travel without putting them to sleep. Damaging sites still delays production, and the local population cap still applies.

Get KLA's to fly into enemy structures

  Corncob was designed so that enemy objects can collide with each other.
A useful technique for dealing with a long string of tailing KLA's is to simply trick them into flying into their own structures.
  For example, it is possible even to blow up a generator orb by leading deathballs into it.
  For best results you will need to keep your airspeed up, and make your 180 degree turns be gentle to avoid dispersing the stream.

Destroying strings of KLA's with your runway:

  If you have used all your bombs, and there is a long string of KLA's about to hit you on your landing approach, there is no need to worry. The materials used in the runway will cause the KLA's to detonate prematurely if they get close.
  Even if you have no intention of landing, a low flyby over the runway will cause the trail to explode as they follow your lead over the runway.

Get rid of a KLA stream by dropping a low bomb:

  Certainly you can take out a single KLA which is very close to your tail
simply by dropping a bomb. You aren't required to do any maneuver, just drop the bomb. However, you will frequently find that you have a long string of them behind you, and only three bombs. By flying low, and dropping a bomb, the debris flying up from the explosion can take out almost the whole string as they attempt to fly through the explosion.

Keep your airspeed up around AAA:

  The higher quality AAA will attempt to lead your plane so that the slow moving shells will arrive on target despite the forward motion of your plane. Thus, the higher your airspeed, the more dispersed the enemy shells will be.
  Letting your airspeed drop due to maneuvers is the fastest way to get blown out of the sky by AAA. When you drop your flaps to mix it up with the KLA's is when they will get a bead on you. Eliminate all the top notch AAA before getting into knife-fights with the deathballs.

Stay low to avoid AAA fire:

  Ground-based AAA batteries cannot fire very low to the ground, for fear of taking out their own structures. It is possible to stay under their firing angle by flying extremely close to the ground. The closer you are to the AAA site, the lower you must be to avoid fire.

Stay level with suspended AAA:

  AAA shells are programmed at firing time to explode at a certain

General strategy and flying hints (continued)

altitude. Since the enemy is attempting to kill you, they program each shell to explode when it reaches the altitude of your plane. The exact altitude of detonation is randomly distributed around the target altitude.

  The result is that if you manage to fly level with the AAA, the shells are already at their detonation altitude before they even leave the AAA muzzle! Therefore the shells will almost always detonate before hitting your plane.

Always use bullets and missiles when attacking AAA:

  The bad news is that AAA shells will explode your missiles as you fire on an AAA site. Since the AAA site will be sending a continuous stream of shells at you, then it is unlikely for your missiles to make it to the target.

  The good news is that bullets will explode approaching AAA shells. Therefore the best technique is to let loose a volley of bullets to take out the slow-moving approaching AAA shells. Then send a group of missiles together with more bullets. That way the faster-moving bullets will take out the AAA shells, clearing the way for the powerful missiles.

Never fly with your tail to the AAA:

  AAA works best on zero-deflection targets. If you cannot use your weapons to clear the air of approaching shells, then your best bet is to be flying fast and perpendicular to the AAA. You are most vulnerable when flying directly away from the AAA site. Then the AAA can shoot directly at you with no target-leading or anticipation, and the rate of closure is the slowest to ensure an accurate blast.

Avoid bombing AAA unless desperate:

  Many a pilot has been toasted by his own bomb whilst attempting this. The AAA shells will detonate your bomb in the air. For lower quality AAA, the pilot can sometimes get lucky when the firing rate of the site is fairly low.

Tips for downing flying saucers

Keep up the airspeed:

  When dealing with saucers with AAA, you must be aware of the fact that you will not be able to down the saucer with one pass. Therefore after passing the saucer, it will chew you up as you attempt to fly away in order to make another pass. Your only defense is fly as fast as possible, so that the time in which your tail is turned toward the saucer is minimized.

  In addition, the faster your airspeed, the faster your missiles will reach the target. The launch speed of your missiles affects the acceleration to max velocity in a positive way. With a moving target like a saucer, the faster your ordnance travels the better.

Keep level with the saucer:

General strategy and flying hints (continued)

For the same reason that you should stay level with suspended AAA, you should stay level with AAA fitted saucers. This is especially true when you are pulling away from the saucer.

Try to time the oscillations of the saucers movement:

You will notice that the saucer's evasive maneuvers are very periodic. Therefore if you notice that your missiles are always missing because of the saucer's motion, try shooting where you think the saucer will be when the missiles arrive. If that isn't working, just spread your missiles around the whole area through which the saucer is swinging.

Try to get 'behind' the saucer:

At any moment, all the saucers are traveling in a horizontal direction. You want to approach in the same direction as the saucer is moving. This gives you a slower rate of closure, so that you can get more shots in before you have to peel off to avoid collision. It is possible to down a saucer in one pass using this technique.

Do not fight saucers in close:

Fighting a saucer up close usually ends in disaster. The evasive maneuvers which it is constantly executing make it too unpredictable a target at close range. In addition, there is a great risk of collision. Upon collision with a saucer, your plane becomes a ball of twisted wreckage.

Although collision will take out the saucer, the brass doesn't look at it as an even trade. Perhaps for the desperate, it may be worth suiciding a dumpy plane for a top-ace saucer, though.

Hints for vanquishing ground vehicles:

Keep your distance:

Ground vehicles do much random zipping around. If you are up close, those movements will cause you to go through wild aerodynamic gyrations in order to keep those critters in your sights. This is very likely to end in a fatal stall, since when attacking ground targets you are probably flying too low to recover.

From a prudent distance, however, the cart's erratic movements can be tracked with minimal difficulty and danger. Use the plane's rudder to spray bullets and missiles into a cloud about the vehicle. Shopping carts are not very tough, and in just a few lucky hits they will fall over dead.

Lead the top-notch carts into traps:

General strategy and flying hints (continued)

The top-notch shopping carts will follow your plane with uncanny ability. It is almost impossible to get a clean shot at them because they are always directly underneath your plane.

If they are the variety which do NOT have the flashing antenna, then your best bet is to fly directly over a fence, tick, or other structure in order to lead them to their own destruction.

Climb and shoot:

The high-quality flashing antenna carts can be defeated by climbing high, dropping the flaps, and then come straight down at them with guns blazing. Make sure to break off before you pile into the ground.

Vertical bombing:

Ordinarily, bombs are not effective against the agile carts. However, in the case of the high-quality models, you can use their tenacity against them. Fly exactly straight up. Your pitch gauge should show your plane as being straight up. Then drop three bombs, and remain vertical until the bombs explode. If you are lucky, the shopping cart will have stayed directly below you doing a little dance while the bombs rained down.

Note that if you stall and fall out of your straight up climb before the bombs reach the ground, then the shopping carts will have followed you and moved out of the target area by the time the bombs fall.

Attack shopping cart sites first:

As with KLA's, it is a losing battle to try take out all the carts first, and then the sites. Instead, ignore the carts until all the sites are history. Then have fun tearing up the panicked vehicles.

Tips for survival outside the plane

Get away from the plane after leaving it:

When you are outside your plane, you are very vulnerable to enemy weapons. Contact with a KLA, or AAA shell results in instant death. Since both weapons will attack your plane, not yourself, it is a good idea to get far away from your plane very quickly after ejecting.

Keep moving at all times if there are shopping carts around:

The shopping carts are like the KLA's in that they always head directly toward (despite their erratic motions) you. IE unlike the AAA, they do not lead. Therefore if you run in a straight direction, they will always end up chasing behind you. If you stay put, they will wander in from all directions until you have no direction which is safe.

General strategy and flying hints (continued)

Look down at your feet when jumping platforms:

When hopping from platform to platform, tilt your head down 30 to 40 degrees from the horizontal. Otherwise when you jump, you will not be able to see the platform you are attempting to jump onto. Jump high into
the air, and hold and release the W key to move forward just the right amount to land you on target.

Watch the far edge of the platform you are jumping to. As long as you don't pass over that edge, you will make the jump. Keep that far edge within your view, and you should have no troubles.

Use proper vertical angle with the handgun:

Don't always assume that you should aim for the center of an object when shooting with the handgun. Some targets take more hits with the game aimed high. If you see grey spheres appear where the bullets hit, then the bullets are striking the ground before damaging the target.

Experiment with position if assassin bomb fails:

The object-object collision algorithms used in Corncob were designed for fast-moving objects colliding with stationary targets. Assassin bombs will not always detonate their targets correctly because of the design of the game engine. For instance, if a mind bender is sitting on a force field, then a tricky situation can result. Both the force field and the mind bender can detonate a bomb. Therefore, when the detonate command is given, either the force field or the mindbender can be the cause of the bomb's detonation. If by chance the force field detonates the bomb first, then the mindbender will remain intact through the bomb's detonation.

Which object causes detonation is determined by the exact positioning of the bomb relative to the target and the person. Before collision processing, all objects are sorted by distance from the observer, then objects which are close together in the sorted list are compared for collision.

So if your bomb fails, try the next bomb in a position closer or farther away from the target. Make sure the center of the force field platform is not approximately the same distance as from you to the target, etc.

Use parachute to get extra hopping distance:

Your parachute can be used even when jumping into the air. If you are particularly nervous about a jump, you may pull the chute string at the apogee of your jump, and then float down slowly. The disadvantage is that you will swing for a bit, which can be disorienting.

The Corncob Program Itself

Corncob 3D was written by Kevin Stokes, with a lot of additional programming by George Welch. The program documentation was written by Kevin Stokes. Original music was written by Jim Collymore, A.S.C.A.P. All music in the game is copyright 1986-1991 James L. Collymore, and the use of the three music files for any purposes except for Corncob 3D is strictly illegal. The music is contained in the following files: VOYAGE.ROL, DAYBREAK.ROL, and HEAVENS.ROL. These files cannot be legally
copied or distributed without the author's permission.

The VGA directory (created the first time Corncob 3D is run) contains all the digitized artwork in the game. These photos were extensively retouched using the excellent program MVP Paint, by David Johndrow. You can order your own copy of this extremely useful program from MVP Software.

Corncob 3D was written in assembler and C. The excellent library of routines Spontaneous Assembly was used in developing the program.

This software is based in part on the work of the Independent JPEG Group.

Contacting MVP Software & the Corncob BBS

[Historical DOS documentation - retained for reference]

Contacting MVP Software & the Corncob BBS
  Call our 24-hour BBS (2400,n,8,1) and download the latest shareware version of Corncob on the first call. This is also the best way to get technical help. Call (919) 489-6900.

  To contact MVP Software, call (616) 245-8376 or write us at 1035 Dallas
SE, Grand Rapids, MI 49507-1407. We can be reached on CompuServe at 70363,15.

  Corncob's author can be reached on Prodigy at DWHC72A, through Internet
at kds@phy.duke.edu, and on Genie at K.STOKES1.

Using the Mission Builder

Instructions on how to use Corncob's mission builder.

Copyright 1992 Pie in the Sky Software

Theater files and tower files
Redux integrates the mission builder into the game. Choose Select campaign / world group, then Mission builder. Choose a world to edit; the editor uses your current pilot's saved copy. F5 saves and returns to the menu. A copy of the edited definition and its tower text is exported into custom_worlds inside your Redux save directory. There is no need to run 3.EXE, copy 3UNIV.DAT, or use DOS batch files.

The following description of DEF and TWR structures and the source tables is retained for reference. Internal object names and assembly tables are reproduced as written; they are not commands to type into the native executable.

For every .def file in the DEF subdirectory, there is also a corresponding
.twr file. The .twr file contains intelligence information that the user
will see when he requests it via F2, or at the beginning of the mission. The tower files are just normal text files with 9 formfeeds. (Control L) The body of text between each formfeed is a block of text that the user will see when he starts from that particular airstrip. The easiest way to create your own tower file is to modify an existing tower file, rewriting only the actual mission info, and leaving the airstrip names and formfeeds alone.
Some text editors may not like the formfeeds. The tower files were originally created with MKS vi editor, which handled the formfeeds with no problem.

The start positions for the missions are programmed into the tower files.
Each mission description has a line with the characters Sxyth in it. The numbers following these 5 characters contain the x, y, z, and yaw angle.

To program in your own start coordinates, simply use the Map mode to position the viewpoint where you want the pilot to begin the mission. Hold down the left shift key, and use the keypad arrow keys to change the angles of the view point. When you are satisfied with the position and angle, simply copy the x,y,z and yaw angle numbers off the bottom of the screen.
Then edit the tower file, and insert these numbers after the Sxyth characters. Make sure you leave spaces between the numbers, and don't put any extra spaces at the beginning of the line.
Note that you can only set the yaw angle (compass heading type) of the plane's start. You can't change the initial pitch or roll. IE you can't make the plane start out flying anything but level at the start of a mission. You can only change the compass heading which the user will start out at.

The sun position, air pressure and gravity.

Tower files can start out with a line like the following:

X 6000 300 12 -48 10000 0 1500

Using the Mission Builder (continued)

The X marks that the numbers on this line specify the sun size, position and color. Also is the gravity, air pressure, star flag, and atmosphere extent.

X identifier for theater parameter line.

the 6000 is the angle of the sun in the sky. Range +-32767

300 is the size of the sun Range 10-1000

Color 12 is the color of the sun Color 0-15

-48 is the gravitational attraction Range 0 to -120

10000 is the air density at sea level 6000 is normal atmosphere 10000 max

0 is the star flag. 0 means no stars 1 means stars in sky

1500 is the atmosphere height. 6000 is Earth normal.

How to use the mission builder

To start a fresh universe copy default.def to 3univ.def. Default.def is a universe with just airfields in it.

Put the universe file in a file called 3univ.dat in the current directory.

Start the game with 3 -rs

When sitting on runway in plane, type m to go into map mode. Make sure
the joystick is not drifting, use trim if needed.

Zoom in on the area to be edited with Joystick/Arrow keys, and pg up and pg
down to change height. When close to surface, hit 'b' to enter the mission
builder mode. When in MB mode, the gunsight disappears. IE, you can tell
mission builder apart from map mode by checking for the existence of the gunsight on the screen.
Close to the surface is a height of about 30000-60000 cm. The numbers on
the bottom of the screen read X,Y, and Z in cm. (Z is height)

When in MB mode, the Joystick/Arrow keys will change the viewing direction,
just as when you are an ejected pilot. Just like an ejected pilot, the numpad keys '5', and 'ins' will move you forward or backward in the direction
which you are facing. The constraint that you remain on the ground is however
lifted.
When slewing around in this method, the non-keypad number keys will change

Using the Mission Builder (continued)

your walking rate. IE '1' makes you move slowly, '8' makes you move quickly.
Caps lock will instantly level you out, so you are looking at the horizon.
This is usefully when you want to move in a perfectly level direction.

  Now to select an object for moving/deletion/changing, you must bring up
the 3d cursor. Do this by tapping the left shift key. A rotating ball of spheres should appear in front of you. If you move about, the cursor
remains in its position of creation. Do not move too far away from the cursor,
or the program will become confused because it cannot operate on a cursor
which is in far object memory. Be careful when using pgdn especially. If
you are going to move a far distance, turn off the cursor with the left control key first.

  If you wish to move far away, tap the control key to make the cursor disappear. Also, you may return to the map mode with the cursor either lit
or unlit, and move around in map mode, and then back to builder mode via the
'b' key.

  The cursor can be moved with the Joystick/numpad arrow keys, if the alt
key is held down while doing so. The pgup and Pgdn will move the cursor up and down, however your viewpoint will also change height at the same time.
(IE, when you move the cursor up or down by holding down the left alt, and
pgdn or pgup, the cursor will remain at a fixed position relative to you, but
you AND the cursor will move up or down.)

  When you turned on the cursor, note that the numbers came back onto the bottom of the screen. These are no longer the viewing coordinates, these are the coordinates of the cursor. The number 21 is the object type. When the cursor is displayed, the number is always 21. The last number in the list is the object quality. This can be the wickedness of an AAA, or in the case of a Headquarters orb, this defines the mission objective. As with many types of objects, the quality of the cursor is ignored.

  So anyway, when you are near the ground, and the cursor is on, tap the 'c' key to create an object. The default object is a headquarters orb. It should create an orb at the cursor position which is big and blinking.
  Each type of object has an object number. Object number 0 is a generator
orb.   Object number 1 is a mind bender, object number 2 is a plane. A list
of objects and their type numbers is included with this doc.

  Use the non-keypad += keys to cycle through the available objects.

Using the Mission Builder (continued)

The
current object should be sitting there blinking slowly. Each time you
press the non-keypad + key, the object will metamorph to the next object
type. IE the HQ orb will turn into a mindbender. As you cycle through
all the different objects, they will appear at their default height
above
the ground. This can be a pain sometimes, especially when creating
missions
high in the air.

  As long as the object is blinking, you may move it around with the
arrow
keys just like the cursor. To make the keypad arrow keys operate on the
object instead of moving yourself, just hold down the left ALT key while
you press the keypad arrow keys. The number keys will affect the motion
rate
just like when moving about. Be careful not to lose track of where the
object is. If the object doesn't appear to moving in the direction you
are requesting, try tapping the non-keypad '1' key, to set the speed to
slow.
  Note that objects will always move in a horizontal plane when moving
them
with the arrow keys. An object's height can only be changed with the
pgup and pgdn keys.

  A blinking object will be deleted if the 'd' key is pressed. IE the
current object which you have just created (or selected) can be removed
from the universe with the 'd' key.

  To make your blinking object a permanent object, press the left shift
key. The currently selected object will stop blinking, and is now a
permanent object just like any other object in the universe.

  If you wish to make changes to existing objects, then you must first
make that object become the current object. (IE make the thing blink)
  To select your object, move the cursor close to the center of the
object, then hit the space bar.   The space selects the object which is
closest to the cursor.   BEWARE: Do not use the space bar if there are
no objects even close to the cursor, you may get kicked out of the
program with a critical error.
  Make sure the correct object is blinking, if it is not, then probably
some other object (which perhaps you can't see from where your looking)
is selected, and all changes you make will be happening to the poor
hapless
object which you have unwittingly selected.

  Objects can be rotated in place by holding down the 'a' key, and then
using the arrowkeys/Joystick to rotate the object. To lock the object
to the nearest 90 degree plane, press the appropriate key.
NOTE: All objects can be rotated, but non-circular objects such as
forcefields should NOT be rotated because the rectangle of collision
of the forcefield will not be rotated with it's image.

  Some of the objects have a 'special bit' which can be set with the
mission builder. For example, if a generator orb has its special bit
set, then it will be a gravity orb. If a shopping cart has its special
bit set, then it can only be destroyed by a human weapon, and will never
die via collision with another object.

Using the Mission Builder (continued)

This special bit is set with the F9 key. When an object's special bit is set, the object type number on the screen bottom will turn purple.

Like the special bit, the F10 key will set an object's complete bit. The complete bit matters in only a few objects. This is the bit which keeps track of whether or not a mission objective has been accomplished. If you set the complete bit of an object, then the object type number at the bottom of the screen will flash.

One use of the complete bit is to make portals which would ordinarily count as a mission objective, not count. IE say you wanted to make a mission where you had to fly by a bunch of portals on the way to your real mission objective. If you just create the portals, then all of them would count as mission objectives. IF you set their complete bits ahead of time, then they will count has having been already flown through.

Zlocking

Often one wants to make many objects all be created at the same height. This can be accomplished with zlocking. Simply put the first of these objects at the desired height. Then while that object is still selected, press the left alt and the 'h' key. This records the present z coordinate into zlock-height register. Now create your next object. To force it to be the same height as the zlock-height, simply press 'h'. The z coordinate number at the bottom of the screen should turn black, and the object should be the same height as the last one.

The z coordinate number should turn black whenever the present selected object's z coordinate matches those of the zlock-height register.

Notes on where to build

Each theater is made of an 8x8 grid of tiles. These square tiles each can have a maximum of 100 or so objects. This limitation should never be approached, as this will create a critical error, thus propelling you swiftly out of the program will all your changes lost.

The tile boundaries are mapped out with survey markers which can be seen on the ground, and also show up in map mode as a grid of yellow or black dots.

During game play, only the objects in the nearest 4 tiles will be displayed. Therefore you can see that when the user is flying in the middle of a tile is always when one tile is swapped in, and another out. In other words, the computer will have to do some extra work to swap out a tile, and read in a new one when you cross the middle of a tile. For this reason, it is a good idea not to put missions in the centers of tiles. If you do, then the game play will be continually interrupted for tile swapping during the acrobatics.

Note that this applies both the up down, and left right tile centers. Picture a square tile with a '+' sign on it, where the plus sign is big enough to cover the whole tile. The imaginary lines of this plus sign show the areas where tile swapping will occur. Therefore, your missions should always be closer to the corners of the tile, then the tile

centers.
  Notice that the frame rate will suffer dramatically if you try to put too much stuff close together on neighboring tiles. For example, you might want to make a mega-mission at the intersection of 4 tiles.
  You could put up to 100 objects in each tile, and still have it all fit together as one mission. However, in reality, so many objects to be calculated and drawn would lead to an absolutely terrible frame rate, and an unplayable mission.
  You must also be aware that even if you don't have 100 objects in a tile, you could still end up with a critical error during game play
  The reason is that passing KLA's, AA shells, and shopping carts all count as objects in that tile, even if they were created in one tile, and migrated to the other.
  If you look at the default theaters that came with the game, you will notice that most all of the bigger missions are made to be at corners of tiles, and well away from the airstrips. This is simply because of the need to keep the frame rate up. This is especially important for those players with machines slower than 25Mhz.
  You will notice that the frame rate is not so great at Sweetly Bending in Flatland because the missions were too close to the airstrip. This makes landing there difficult.

Notes on what to build
  Usually start with a generator orb. Set the quality to corresponding to what you want the mission object to be.
  Keep in mind that the many players will be bored with just a bunch of KLA's and deathballs. What makes a mission fun is not just a challenge, but a challenge that takes strategy to lead to victory. Examples: A mission where there are shells beneath an orb, so that if the play tries to take down the forcefields first by destroying the orb, then the shards that fly off the orb will trigger the shells, thus bring a whole slew of enemies upon the unwary player.
  How about putting shells down on the ground where they wouldn't normally
be in danger of triggering, but then but some high quality vehicles on the
outskirts, where they will follow the plane in, and set off the shells even if the flyer doesn't shoot the shells himself.
  The force fields open up a whole lot of possible structures that the pilot must navigate to successfully complete the mission. Beware of overuse of the solid or solid-shimmering fields, as they soak up the cpu time to draw. Use the wire frame versions where possible.

Mission builder: why is it doing that?
> Why does it seem impossible to get the height of objects right when >they are pretty high up?
   The mission builder makes the pgup and pgdn keys have an effect proportional to the height of the object. When the object is relatively high up, the vertical steps are huge. This can be gotten around via the 's' key. When this key is depressed, the pgup and pgdn keys will always move a relatively small distance (1000 cm) regardless of the object's height.

> Why are the objects always the wrong color, and sometimes the wrong >shape when I go into the mission builder or map?
  The way the program is written makes the qualities of objects never matter when in the mission builder. This is because the code which makes the object the correct color etc is the same code which makes the

object move, shoot etc. Therefore if that part of the code were turned on during the mission builder, then the objects would begin to move and fire as soon as they were created. So unfortunately, you will have to trust the quality number on the bottom of the screen to see the actual
quality of the object.

> I worked for two hours on a mission, and when I tried to get out of
>the mission builder mode when finished, I got critical error 70, and
>lost all the changes.
   The mission builder is not perfect, and I typically only make small changes, and then exit and back up the 3univ.dat file. This process is relatively quick, and all the default theaters which came with the game were created with the same mission builder which is available to you. Occasionally you will lose stuff, but if you never stay too long in the MB, and don't keep skipping back and forth between map and MB mode, you should never lose too much.

>What are the things to avoid doing in the mission builder?
  At any one during game play or during mission creation, the computer can only deal with the objects which are relatively close. When an object
gets too far away to the observer, it is placed into the 'Far object table'.
  Objects in the 'Far object table' are inactive, and are stored in compressed form.
  The mission builder can operate on near objects only. Far away objects
cannot be processed by the builder.
  It is VERY important that you never ask the builder to act on an object
which is out of the near range. This will cause unpredictable behavior, and critical error codes.
  Some examples of causes of this:

  Say you create an object on the ground, and then zoom your viewpoint up
to 40,000 feet. Then you start moving the object around on the ground. If
you move that object too far away, you will trap your newly created object
in limbo, and you will be risking a critical error.
  Try to stay near your current (blinking) object at all times.

  Say you start to build a mission, and then decide to scrap it. Deleting
the last object will get you a critical error, since there are no objects
at all to process.

  Say you are using the keypad + and - keys to cycle through the near objects.
You get impatient, and hit the - key repeatedly very fast. If you cycle through
all the near objects, then the poor mission build tries to pick up a far object
and trouble ensues.

Using the Mission Builder (continued)

All these limitations may make the mission builder seem like it isn't very useful, however with the proper habits, you can use the MB continuously for days without a mishap. This very program was used to create all 150 of the available missions in Corncob.

Just remember to only make one mission at a time, and make backups of the theater to another filename after every mission. Don't flip back and forth between map and mission builder mode. Start the program, make your changes, and exit back to DOS between each change. Take care not to move too far away from the current blinking object, and never travel any great distance withing the mission builder. If you want to move somewhere else in the theater, exit the program completely, and start it up again.

To exit the mission builder, you can hit escape or 'm' to return to map mode, then escape or space bar to return to flying mode.

Do not fly after returning from the mission builder, just exit, then copy 3univ.dat to carl.dat or whatever. Its a good idea to make backups in case of unexpected critical errors.

```
; The following is taken from the sources of Corncob.  This is the
; definition tables for the objects.  Use these tables to get the
; object number of what you would like to create, and what quality
; you would like to assign that object.

                    ; number after the semicolon is
                    ; the object type number

templatetable   dw  @dataseg:objhq  ;0  Headquarters orb
    dw  @dataseg:objstet    ;1  mindbender
    dw  @dataseg:obj5da     ;2  plane  qual2=dumpy  qual3=Stealth
    dw  @dataseg:objrdr     ;3  radar
    dw  @dataseg:objtw      ;4  radar base
    dw  @dataseg:roadpiece  ;5  Runway piece  qual2= no stripes
    dw  @dataseg:tower      ;6  Control tower
    dw  @dataseg:objman ;7    Ground crew
    dw  @dataseg:objshard2  ;8
    dw  @dataseg:objshard   ;9
    dw  @dataseg:objincend  ;10
    dw  @dataseg:objdbst    ;11 KLA site
    dw  @dataseg:objvech    ;12 Vehicle  spec bit= bullet only
    dw  @dataseg:objvest    ;13 Vehicle site
    dw  @dataseg:objwallz1  ;14 Force fields
    dw  @dataseg:objwallx   ;15        "
    dw  @dataseg:objwally   ;16
    dw  @dataseg:objwallz   ;17
    dw  @dataseg:fltsm1puff ;18
    dw  @dataseg:fltsm1boom ;19
    dw  @dataseg:fltsm1bomb ;20
    dw  @dataseg:fltsm1mis  ;21
    dw  @dataseg:fltsm1aashell  ;22
```

Using the Mission Builder (continued)

```
    dw  @dataseg:fltsm1aa   ;23  AAA site
    dw  @dataseg:fltsm1dball    ;24  KLA
    dw  @dataseg:fltsm1     ;25
    dw  @dataseg:fltsm2     ;26
    dw  @dataseg:gms        ;27   Fuel dump
    dw  @dataseg:obj5d      ;28   Do not use!
    dw  @dataseg:objprop    ;29   Propeller
    dw  @dataseg:bullets    ;30
    dw  @dataseg:objsurv    ;31   Survey marker
    dw  @dataseg:crtr       ;32   Crater, qual 1= permanent
    dw  @dataseg:smgms      ;33   Destroyed tick
    dw  @dataseg:objhut     ;34   Tower doorway
    dw  @dataseg:objroadend ;35   runway endpieces
    dw  @dataseg:objroadend1    ;36
    dw  @dataseg:objfn1     ;37   Fences
    dw  @dataseg:objfn1a    ;38
    dw  @dataseg:objfn1b    ;39
    dw  @dataseg:objbb      ;40   Barrage Balloon
    dw  @dataseg:objbbsm    ;41   Bumble Bee
    dw  @dataseg:objmort    ;42   Mortar
    dw  @dataseg:objdhq     ;43   Blown up HQ
    dw  @dataseg:objlwallz1 ;44   Line-type forcefields
    dw  @dataseg:objlwallx  ;45
    dw  @dataseg:objlwally  ;46
    dw  @dataseg:objlwallz  ;47
    dw  @dataseg:objrv      ;48  Rescue Van
    dw  @dataseg:objshell   ;49  Shell
    dw  @dataseg:objbmr     ;50  Saucer
    dw  @dataseg:objbmrst   ;51  Saucer Site
    dw  @dataseg:objtet     ;52  Tet
    dw  @dataseg:objbmrmo   ;53  Saucer that is a mission objective
    dw  @dataseg:objslwallx ;54  Smaller line-type forcefields
    dw  @dataseg:objslwally ;55
    dw  @dataseg:objslwallz ;56
    dw  @dataseg:objal      ;57  Alien
    dw  @dataseg:objxp      ;58  Portal
    dw  @dataseg:objstep    ; 59 Small step forcefield
    dw  @dataseg:objpstep   ; 60 Small step forcefield Filled

    ; the following are the definitions of the different qualities
    ; of portals.  Ex: qual 3 takes us to base, but damages us.

xpqtbl      dw  4
quality 1   dw  11,11+10h,1,0       ; takes us back to base
quality 2   dw  3,9+10h,255,0       ; takes us to random place
quality 3   dw  12,11+10h,1,1       ; takes to base, but damages us
quality 4   dw  0,11+10h,1,-1       ; kills us instantly
quality 5   dw  11,11+10h,101h,0    ; takes us to base, fulfils mis
quality 6   dw  11,11,1,1       ; takes to base, but no flashes
quality 7   dw  7,11+10h,0,2        ; repairs plane, doesn't xpose
quality 8   dw  10,3,100h,0     ; doesn't move us, miss obj

    ; the following are the mission objectives of the different
    ; headquarter orbs.   Example: a quality 1 orb counts as a mission
        ; objective when you blow it up, a quality 4 orb will count a
mission
        ; objective when all the vehicles and vehicle sites are gone.
```

Using the Mission Builder (continued)

```
hqtbl   dw  2
1)  dw  0,@codeseg:killme
2)  dw  6,@codeseg:killticks
3)  dw  7,@codeseg:killdbaa
4)  dw  3,@codeseg:killvechs
5)  dw  9,@codeseg:killmetough  ; must suicide plane to destroy
6)  dw  13,@codeseg:killbmrst
7)  dw  4,@codeseg:killstets     ; kill all mindbenders
8)  dw  14,@codeseg:killal       ; kill aliens

; the following shows what qualities of shell will turn into what.
; I don't think qual=8 works

shellqtbl   dw  3       ; table of qualities for dballsites
1)      dw  13,0,6      ; vech stite
2)      dw  42,6,7      ; mortar
3)      dw  11,7,3      ; dball site
4)      dw  23,3,9      ; aa site
5)      dw  40,9,13     ; barrage balloon
6)      dw  41,13,11        ; bee swarm
7)      dw  27,11,0     ; a tick
8)      dw  23,2,2      ; 0= random

; The following is a table of various properties of objects.  You cannot
; change any of these settings, it is just here for reference.
;  The numbers in the table are sums of attribute bits.  These number
are
; powers of two.  Example 512=2^9, so +512 means that this object will
; capture you if you touch it.
;
; Example: object 10  (incendiary from bomb, or mortar)  2^0=1
; 1=disappears when goes off into the distance, not saved in far tile
ram
; 2=Skip over this object when in mission builder, who'd want to create
this?
; 4=If you fly through it, object is terminated  4=bit 2  IE 2^2=4

;1  ; bit 0 means do not xfer to medium table
;2  ; bit 1 means skip over when doing mission builder creates
;4  ; bit 2 means terminate object if plane hits it
;8  ; bit 3 means it is forcefield type object
;16 ; bit 4 means it is a runway type object
;32 ; bit 5 means object is a portal
;64 ; bit 6 means object is standonable
;128    ; bit 7 means object kills you
;256    ; bit 8 means object walk height is collision + z coord
;512    ; bit 9 means object captures you

extendedtypetbl dw  0+4 ;0
    dw  0   ;1
    dw  0+4     ;2
    dw  0+4     ;3
    dw  0+4     ;4
    dw  0+4+16+64   ;5
    dw  0+4+64+256      ;6
    dw  0+4 ;7
    dw  0+2 ;8
    dw  0+2 ;9
```

Using the Mission Builder (continued)

```
    dw  1+2+4   ;10
    dw  0+4 ;11
    dw  4+512   ;12
    dw  0   ;13
    dw  0+4+8+64    ;14
    dw  0+4+8   ;15
    dw  0+4+8   ;16
    dw  0+4+8+64    ;17
    dw  1+2 ;18
    dw  1+2 ;19
    dw  1+2 ;20
    dw  1+2 ;21
    dw  1+2+128 ;22
    dw  0+4 ;23
    dw  128 ;24
    dw  1+2     ;25
    dw  1+2     ;26
    dw  0+4+64+256      ;27
    dw  0+2     ;28 skip over obj5d, so cursor don't dis
    dw  1   ;29
    dw  1+2 ;30
    dw  0   ;31
    dw  0+128       ;32
    dw  0       ;33

    dw  0       ;34
    dw  0+16+64     ;35
    dw  0+16+64     ;36
    dw  0+4     ;37
    dw  0+4     ;38
    dw  0+4     ;39
    dw  0+128       ;40
    dw  0+128       ;41
    dw  0+4     ;42
    dw  0       ;43
    dw  0+4+8+64    ;44
    dw  0+4+8   ;45
    dw  0+4+8   ;46
    dw  0+4+8+64    ;47
    dw  1+4     ;48
    dw  0       ;49
    dw  4+64+256        ;50
    dw  0+64+256        ;51
    dw  0       ;52
    dw  4+64+256        ;53
    dw  0+4+8   ;54
    dw  0+4+8   ;55
    dw  0+4+8+64    ;56
    dw  0+512       ;57
    dw  0+4+20h     ;58
    dw  0+4+8+64    ;59
    dw  0+4+8+64    ;60

; Special bit definitions:, hit F9 to set an object's special bit
; The object's type number will turn purple to indicate that it's
special
; bit is set.
;
```

Using the Mission Builder (continued)

```
;  1.) Orbs with the special bit set will be gravity-attraction orbs.
;  2.) Vehicles with the special bit set will be destructible with
;        bullets or bombs ONLY, objects will last forever.  The antennas
;        of these vehicles will flash.
;  3.)  Mindbenders with this bit set will be susceptible to damage only
from
;         assassin bombs, and are invulnerable to plane-dropped bombs
;  4.)  Ground Crew with this special bit set are spies.  They wear
black and
;        their mission objectives are complete when you go near them.
;

; Complete bit (F10 toggle), is useful with the following objects
;  When the complete bit is set, the objects type number blinks
white/black.
;
;  1.) Orbs which you need, but you don't want them to be mission
objectives
;        in any way.  If you make one, but preset it's complete bit,
then
;        the player will not need to accomplish that orb's objective to
;        finish a theater.  This is useful when you want an orb to power
;       the local force fields, but you don't want the pilot to make the
;       mission to be trivial because he can just blow the orb to drop
the
;       force fields.  The solution is to make a kill-me-tough orb with
it's
;       complete bit already set.  Then the pilot can't kill it without
;       sacrificing a plane, but the player doesn't NEED to kill that
orb
;       in order to complete the theater.
;
;  2.)  Portals which may be desired, but not count as mission
objectives.
;       Just preset their complete bits in order to make portals which
the
;       the player may use, but do not have to be flown through to
satisfy
;       an MO.
;
;  3.)  Mission objective-type saucers where you want one to stay put
like
;         a mission objective saucer, but you don't want it to count as
;         an MO.
```